Dynamic Job Market Signaling and Optimal Taxation

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Abstract

How should workers' dynamic reputation-building efforts affect the design of optimal taxes? To answer this question, this paper first presents a model in which job histories play a crucial role in transmitting information about workers' productivity. Workers' reputational benefits from overworking to signal higher productivity generate dynamic "rat race" externalities. The paper then provides general optimal taxation formulas, written as the composition of a redistributive and a corrective component. The latter depends critically on dynamic labor wedges: the difference between the marginal contribution to output and the sum of current and future benefits the worker receives from supplying one extra unit of labor. Empirically, evidence from the Health and Retirement Study suggests a substantial and positive corrective tax component, especially among high-income earners. JEL codes: D82, H21, H23, J20.

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1 Introduction

Dynamic informational imperfections shape workers' incentives to exert effort, affect the income distribution, and, thus, impact the tradeoffs between efficiency and redistribution from raising taxes. At first, foreseeing higher salaries in the future, workers exert a great deal of effort even though their current salaries are small. As workers accumulate experience, employers become able to recognize who the more productive workers are and start paying them differentially (Farber and Gibbons, 1996; Altonji and Pierret, 2001; Lange, 2007). However, employers may still be unable to distinguish between workers of different productivity levels perfectly, and those workers may end up receiving the same wages.

How workers' incentives are shaped and how income is distributed are also primary considerations in income taxation. However, standard benchmark taxation models often ignore dynamic informational asymmetries, either adopting a stylized, frictionless view of labor markets where workers' salaries are equal to their marginal products or adopting a static perspective on informational imperfections. This paper fills this gap, showing how dynamic informational imperfections affect the equity and efficiency tradeoffs the government faces when setting taxes. To achieve this goal, this paper i) builds a dynamic signaling model of labor markets, ii) derives optimal taxation formulas in terms of sufficient statics, and iii) empirically investigates the magnitude of information imperfections in labor markets and their quantitative implications. Putting these three parts together, this paper concludes that taking into account informational imperfections lowers the cost of redistributing income, pushing toward higher marginal income taxes, especially at the top of the income distribution. Moreover, there can be pure efficiency gains from introducing age-dependent taxes, which would be higher for young workers and would correct intertemporal distortions.

This paper begins by modifying a standard dynamic model of labor supply and demand to address the connection between information transmission in labor markets, incentives, taxes, and inequality. Two modifications are introduced. First, while firms can easily see and pay for the execution of clearly specified tasks or deliverables, it is much harder for them to assess the individual contribution of each worker to the firm's total output. Thus, workers are not paid according to their productivities, but according to the firms' assessments of the value of workers' contributions to output. Second, firms assess the productivity of workers and the value of their deliverables by looking at resumes, which parsimoniously summarize the history of deliverables the workers have produced so far in their careers. When deciding how much effort to exert, workers in the model are aware that working hard brings dual benefits: it generates larger payments today, and it also establishes workers' reputations, signaling their productivity to employers.

This paper then describes the structure of optimal taxes in terms of sufficient statistics, contributing to a growing literature on taxation with imperfect labor markets (Andersson, 1996; Rothschild and Scheuer, 2013; Stantcheva, 2014; Bastani et al., 2015; da Costa and Maestri, 2019; Craig, 2020; Doligalski et al., 2023). I provide general optimal taxation formulas that account for labor market imperfections, written in terms of sufficient statistics. Intuitively, optimal taxes can be thought of in two steps. First, they correct the informational inefficiencies by making sure that, for any extra unit of effort, the marginal benefits to workers are equal to their marginal product of labor. Then, simultaneously, redistributive taxes are imposed on top of these taxes following standard redistributive formulas, where the costs and benefits of redistribution are expressed in terms of welfare weights, compensated and income elasticities, and the shape of the lifetime income distribution.[1] Similar to Scheuer and Werning's (2017) analysis of superstar effects, this formulation allows us to turn on and off the background dynamic signaling effects while holding the distribution of income and other observable moments fixed, and infer the policy implications of taking into account the dynamic signaling effects. By performing that exercise, we will conclude that the costs of redistributing income are lower than we previously thought. Moreover, there may be pure efficiency gains from taxing young workers and correcting intertemporal distortions that arise from disproportionally large reputational gains from increased effort.

To illustrate the patterns for wages and the returns to experience, as well as the nature of distortions that dynamic labor market signaling generates, two polar benchmark examples are presented. In the first polar case, resumes are summarized by the cumulative discounted sum of deliverables a worker has supplied so far, or its "length." I show that in this case, there are no intertemporal distortions, salaries are increasing over time for each worker, and there are high-powered incentives to exert effort, that is, those incentives are stronger than if there were no information imperfections. There are no intertemporal distortions because the compensation for deliverables is independent of their timing. As workers accumulate more experience, they are progressively bundled together with workers who have longer resumes, and eventually, receive more than their marginal productivity. In the second polar case, resumes are summarized by the amount of deliverables each worker has generated per unit of time, that is, the "pace" under which each worker has been supplying deliverables. I show that there can be separation: workers of different types would supply labor at different paces, with more productive workers supplying work at higher paces. Because of separation, firms offer salaries that, in equilibrium, are equal to the marginal productivities, and therefore for each worker, salaries are constant over time. However, that does not imply there are no distortions. In fact, there is another source of benefits workers receive from exerting more effort, which comes from establishing their

[1] Moreover, these generalized formulas apply independently of the model of competition with imperfect information or of the source of the friction that makes workers receive salaries that are not their marginal products, as long as pre-tax earnings map one-to-one to the supply of the supply of deliverables.

reputations and increasing the payments they will receive for future deliverables. Dynamic signaling, in this case, also generates high-powered incentives, but these incentives decline throughout the lifetime of the worker, and approach efficient, undistorted levels toward the end of their careers.

This paper also studies the welfare impact and optimal tax response to recent technological changes that are reshaping how workers are monitored and screened.[2] Building on Stantcheva (2014), it is shown that if redistribution toward the poorer is valued by society, society is hurt whenever there is less information asymmetry because it becomes easier for high-productivity workers to separate themselves from lower-productivity workers. Reducing informational asymmetries tightens incentive constraints and undoes part of the implicit cross-subsidies that happen in labor markets from high-productivity workers toward the lower-productivity workers. Moreover, the elimination of informational frictions in general has an ambiguous effect on taxes, but in simple cases, it i) pushes toward lower marginal corrective taxes, ii) pushes toward higher marginal redistributive taxes, and iii) the first effect dominates the second.

The last set of contributions of this paper consists of empirical results about the presence of information asymmetries, and the extent that they matter for taxation. In that sense, to inform our discussion on redistributive and corrective taxes, the key statistics we need to estimate are dynamic labor wedges, that is, the difference between the marginal productivity of workers and the sum of salaries and future salary increases that results from exerting extra effort. While estimating dynamic labor wedges goes beyond the scope of this paper, this paper estimates static labor wedges, that is, the ratio of marginal productivities over salaries. Surprisingly, in light of the results from our first polar case, where resumes are summarized by their "length," the static labor wedge for the last unit of labor a worker supplies is in fact a sufficient statistic for the dynamic labor wedges, and it offers all the estimates for the size of the Pigouvian taxes on lifetime income that are necessary to correct for the dynamic "rat race." Moreover, the other polar example, where resumes are summarized by their "pace," tells us that incentives are even stronger at earlier moments in the career of workers and suggests that the average Pigouvian correction should be even larger once we take into account additional intertemporal distortions.

To estimate labor wedges, this paper develops empirical strategies that rely on tax changes as a source of exogenous variation in wages and that use data from the Health and Retirement Study survey. We adapt results from the literature that has quantified the degree of adverse selection in insurance markets by leveraging exogenous variation in prices (as in Einav et al. (2010); Einav and Finkelstein (2011); Cabral et al. (2022)). In the context of imperfect information and career concerns in labor markets, the key idea is

[2]These changes appear in the form of increasing availability of data and new tools to analyze it (Chalfin et al., 2016; Autor, 2019; Acemoglu et al., 2020; Bales and Stone, 2020), changes in task composition of jobs from the automation of routine tasks (Autor et al., 2003), and the advent of "new work" (Autor, 2019).

that with a source of exogenous variation in wages for a specific labor contract, one can non-parametrically trace the effects of selection by looking at average productivities of workers who accept the contract as a function of wages.

To measure expected productivities in labor markets, two complementary approaches are adopted. The first approach leverages the Health and Retirement Study data on cognitive measures, treating them as proxies for productivities. We find that the pool of individuals who keep working after a tax increase has better cognitive measures (as measured before the tax change), in line with the idea that the changes in pre-tax salaries induced by tax changes are due to selection, that is, changes in the composition of the pool of individuals who are still working. This effect is also larger at the top of the income distribution, in line with the idea that those informational imperfections are more pronounced for high-paying occupations. The second approach is to treat hourly salaries as the average marginal productivity of workers. Under this assumption, by observing individual salary changes before and after a tax change, and by also observing the number of people who dropped out of the labor force in response to the tax change, we can infer the productivity of those who were originally almost indifferent between staying and dropping of the labor force. Applying this strategy suggests that for an average worker, the Pigouvian component of taxes is of the order of 5%, while for high earners, it ranges from 10% to as high as 60%. Since we can think of optimal taxes as the composition of a Pigouvian and a Mirrleesian component, the fact that the Pigouvian component is so high implies that the redistributive component of taxes is relatively small. Therefore, by taking into account the role of imperfect information in labor markets, we may conclude that the current tax system is less redistributive than we previously thought.

Related Literature The taxation results in this paper build on the optimal taxation literature that goes back to the seminal contributions of Mirrlees (1971); Diamond (1998); and Saez (2001). More precisely, this paper contributes to a growing literature on optimal taxation with richer models of labor markets. The literature has analyzed the cases of educational signaling with complete (Andersson, 1996) or limited instruments (Craig, 2020), competitive screening (Stantcheva, 2014; Bastani et al., 2015), monopsonistic screening and market power (Hariton and Piaser, 2007; da Costa and Maestri, 2019; Hummel, 2021), moral hazard (Doligalski et al., 2023), multidimensional skills (Rothschild and Scheuer, 2013; Ales et al., 2015), and rent-seeking (Piketty et al., 2014; Rothschild and Scheuer, 2016). However, it has not analyzed the case of dynamic job market screening, which is the focus of this paper. Methodologically, the closest papers are Scheuer and Werning (2016; 2017), who show that standard optimal taxation formulas apply quite generally, including a broad range of models where wages are endogenous. Relative to these papers, this paper adds further generality to optimal taxation formulas, enriching them to cover situations where labor market frictions introduce labor market distortions.

In a similar way to the "principle of targeting" (Sandmo, 1975; Dixit, 1985; Kopczuk, 2003), the generalized taxation formulas in this paper hold when there are additional labor market inefficiencies, independently of their nature, as long as the planner can infer labor supply choices from income histories. In this paper, the results on welfare generalize results from Stantcheva (2014), who draws welfare comparisons between economies with "double adverse selection" – in the form of non-linear screening as in Miyazaki (1977) – and economies where firms know the productivity of workers. This paper extends the comparison to more general informational frictions, and to intermediate degrees of informational frictions.

This paper also contributes to the theoretical literature on imperfect information in labor markets. We add a normative perspective to the question of dynamic information transmission in labor market. The dynamic signaling model here combines elements from static models of the "rat race" as in Akerlof (1976) and Miyazaki (1977), and in Stantcheva (2014), with elements from dynamic models of career concerns as in Holmström (1999); Gibbons and Murphy (1992); Bonatti and Hörner (2017); Cisternas (2018), and Hörner and Lambert (2021) among others. The assumption in this paper, that firms see and pay for the execution of observable tasks, or deliverables, borrows from static competitive screening models of the labor markets as in Miyazaki (1977). As in that setup, the quality of the deliverable each worker provides is not observed, and thus individual contributions to firm output are not observed. Firms can see the average output of workers but not individual contributions to output, as in a team production setting (Alchian and Demsetz, 1972). In Holmström (1999), workers' reputations depend on history of output itself. In this paper, reputations depend instead on the history of completion of deliverables, and are summarized by resumes. Furthermore, the idea that firms see a simple public signal builds on the motivational rating setup from Hörner and Lambert (2021).

A more empirically focused literature has looked at how firms learn about the productivity of workers and whether there are information asymmetries in labor markets (Jovanovic, 1979; Farber and Gibbons, 1996; Acemoglu and Pischke, 1998, 1999; Altonji and Pierret, 2001; Lange, 2007; Kahn and Lange, 2014; Cella et al., 2017; Aryal et al., 2019). This paper provides new evidence on informational asymmetries in labor markets, in particular for workers in later stages of their careers, complementing structural estimation results from Kahn and Lange (2014). Additionally, a large literature has analyzed dynamic labor supply decisions, human capital accumulation and on-the-job learning (Heckman, 1976; Eckstein and Wolpin, 1989; Shaw, 1989; Altuğ and Miller, 1998; Keane and Wolpin, 2001; Imai and Keane, 2004; Altonji et al., 2013). One key insight from that literature is that workers would anticipate the impact of their labor supply decisions on their future salaries. The same effect is present in this paper, where the dependence of future salaries on current effort is alternatively explained by signaling.

2 Signaling-by-doing model

This paper adapts a standard dynamic model of labor supply, demand, and taxation, by adding the constraint that firms have limited information over the workers' productivities and can only contract based on a subset of the observed activities workers perform.[3] We focus on one specialized version of the general model that allows us to derive clear comparative statics and simple optimal tax formulas, while retaining the essential economic assumptions that describe dynamic job market signaling. Employers will see a worker's resume – a single scalar that summarizes the history of completion of deliverables from a worker at each point in their career – and will pay workers for the execution of these deliverables. Workers, aware of how their resumes will be read, will choose their labor supply balancing costs and benefits, which include current wages and future wage increases. Appendix B.2, will consider several departures from this simple model, including human capital accumulation and richer signal structures.

2.1 Preferences and Technology

The household block of the model consists of a continuum of overlapping generations of workers with finite lives. These workers have arbitrary preferences over labor and consumption flows, and are forward-looking: they understand that their labor supply choices can affect the information firms will have about them in the future. The production block of the model is described by competitive firms with linear production functions.

More formally, workers in each cohort are indexed by their types θ, which determines their productivity and their preferences. Each worker lives for a continuum of periods that goes from zero to one, where zero corresponds to the time the worker is born and one corresponds to the time their life ends. At each period, workers of all ages coexist. They supply labor ($\tilde{h}$) and consume ($\tilde{c}$) at each period. $\tilde{c}$ denotes the flow of consumption function $\tilde{c} : [0, 1] \mapsto \mathbb{R}^+$, and $\tilde{h}$ denotes the flow of labor supply function $\tilde{h} : [0, 1] \mapsto \mathbb{R}^+$. The histories of consumption and labor supply up to age a are denoted $\tilde{c}^a$ and $\tilde{h}^a$, i.e. $\tilde{c}^a : [0, a] \mapsto \mathbb{R}^+$, and $\tilde{h}^a : [0, a] \mapsto \mathbb{R}^+$, are the $\tilde{c}$ and $\tilde{h}$ functions restricted to the domain $[0, a]$. An individual of type θ has a productivity $v(\theta) > 0$, and production is linear; that is, the flow of production is equal to the product of productivities and the labor supply $\tilde{h}$. Preferences are denoted $U(\tilde{c}, \tilde{h}, \theta)$, where $\tilde{c}$ and $\tilde{h}$ are, respectively, the time-flow of consumption function and the time-flow of labor supply function.

The worker problem is standard: they maximize a utility functional, subject to a lifetime budget constraint, where flows in the future, at age a, are discounted at the rate

[3] The model is a counterpart of the Arrow (1962) learning-by-doing model, where instead of affecting human capital, the completion of deliverables affect only employers' perceptions of the worker productivity. This connection is especially clear in the special case where the resume is defined as the cumulative sum of deliverables a worker has produced. For this reason, we can call it a "signaling-by-doing model."

$q(a)$,[4] as in Equation 1. However, salaries w depend on the information the firms have about the worker $\mathtt{I}(\tilde{h}, a, \theta)$, which will be specified below, but more generally could be a function of the flow of labor supply across all the periods $\tilde{h}$, the type θ and age a of the worker. This captures the possibility that the workers may want to change their labor supply to influence their future salaries, by changing the employers' perceptions of their productivities.

$$V = \max_{\tilde{c}, \tilde{h}} U(\tilde{c}, \tilde{h}, \theta) \ st. \ \int_0^1 q(a)\Big(\tilde{c}(a) - w(\mathtt{I}(\tilde{h}, a, \theta))\tilde{h}(a) - T(\tilde{y}, a)\Big)da \leq 0 \qquad (1)$$

Notice that a standard life cycle labor supply problem features as a special case of this, where wages are equal to the productivity of the worker, and this productivity would be independent of their labor supply decisions. Notice as well that human capital accumulation of the form of on-the-job learning would generate analogous concerns for the worker, with labor supply decisions affecting future wages through real increases in their productivities.

Workers pay taxes $T(\tilde{y}, a)$ on their income flows $\tilde{y}$, which are the product of their wages w and the flow of labor supply $\tilde{h}$. Those taxes can be used to shape incentives and redistribute income. They will be discussed in more detail in Section 4.

2.2 Contracts and Information

Firms are constrained to offer short-term contracts; that is, firms pay workers for the execution of a single deliverable. Firms, cannot individually assess the contribution of each each deliverable from each worker to output, or the firm's profits. That is, firms do not observe workers' types, productivity, nor the quality of the deliverable. Under that assumption, it is worth highlighting that what has been referred to as labor supply $\tilde{h}$ should not necessarily be understood strictly as hours or effort, but as what is referred to in this paper as *deliverables*. That is, $\tilde{h}$ represents the specific piece of the information that firms can pay their workers for.

To assess the quality of the deliverable of the worker, firms observe a signal of their past experience, $\mathtt{I}(\tilde{h}, a, \theta)$. In the simplest case, we can think of that signal as how much labor a worker has supplied so far, which is denoted by $\mathtt{I}(\tilde{h}, a, \theta) = h(a) = \int_0^a \tilde{h}(\tilde{a})q(\tilde{a})d\tilde{a}$, or the length of the resume when the worker is of age a. This case will be discussed in detail in Section 3.1. Alternatively, we can think of that signal as the pace under which a worker has been supplying labor, $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a \frac{\tilde{h}(\tilde{a})}{a} d\tilde{a}$. This case will be discussed in Section 3.2.

[4]These discount rates are assumed to be exogenously given, i.e. there is a technology for transferring resources across periods at rates q, and that these rates are such that budget and resource constraints are well defined; that is, the present value of resources in the economy is finite. While the first assumption is not essential, the fact that the present value of resources is finite is important in guaranteeing that the economy is dynamically efficient.

More generally, we will focus on the formulation where the resume is a scalar of the form $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a \phi(\tilde{a}, a)\tilde{h}(\tilde{a})d\tilde{a}$, that is a weighted sum of past completion of deliverables. The normative properties of this construction will be discussed in Section 4.2, whereas the common general positive properties will be discussed in Appendix B.1. The general idea behind this construction is that the resume is an imperfect measure of the past history of the completion of deliverables and their timing. Together with the overlapping generation structure of the model introduced in this paper, this formulation can be thought of as a parsimonious way of introducing noise in how employers assess what has been done and when. While doing that, it implies that employers may pool together workers from different generations who had a different number of opportunities to work, as long as their past experiences are assessed as equivalent. It allows firms to see and put different weights on past experience; those weights may depend on the current age of the employee as well as the age the employees had when they completed each task.

Underlying the fact that the resume is a signal of how much a worker has produced and how much time it took for the worker to do it, what has been referred to as age a should be interpreted more broadly as the effective time it took for the worker to produce it, that is, *opportunities to work*. While age itself may be an easy piece of information to extract from resumes, it is much harder to verify how much time the worker effectively had to complete all the deliverables, which is arguably the relevant piece of information for assessing how productive a worker is. In particular, for health and family reasons, a worker may not have been able to work as many hours as their age implies. Those obstacles may be hard for employers to infer. Thus, in this construction, firms extract from the resume an imperfect signal about the worker's productivity, how much the worker has produced, and in how much time.

The formulation can also capture in a reduced form different possibilities: firms may be interested in the pace at which the worker has produced deliverables, and use $\phi(\tilde{a}, a) = 1/a$. Firms may want to look at the total experience, as a proxy of human capital, and use $\phi(\tilde{a}, a) = q(\tilde{a})$. It may become hard to verify experiences in the distant past, so that $\phi(\tilde{a}, a) < q(\tilde{a})$. The firms may have all those concerns at the same time, as long as, when put together, they can be summarized by the idea that firms would evaluate the experience through the lens of an index of the form $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a \phi(\tilde{a}, a)\tilde{h}(\tilde{a})d\tilde{a}$. This formulation bypasses the need to introduce explicitly all those elements, and to postulate explicit stochastic processes for shocks for each of those elements and considerations. Not introducing those shocks directly makes the analysis of optimal tax systems tractable, and avoids technical issues arriving from multidimensional screening problems, as well as from failures of the homogeneity assumption on preferences over the timing of consumption and labor supply flows. Appendix B.2.2 and B.2.3 analyze more general cases, where the resume is not a single scalar, including the case where the resume is the whole history of completion of deliverables $\mathtt{I}(\tilde{h}, a, \theta) = \tilde{h}^a$, and thus firms keep track of the (current and

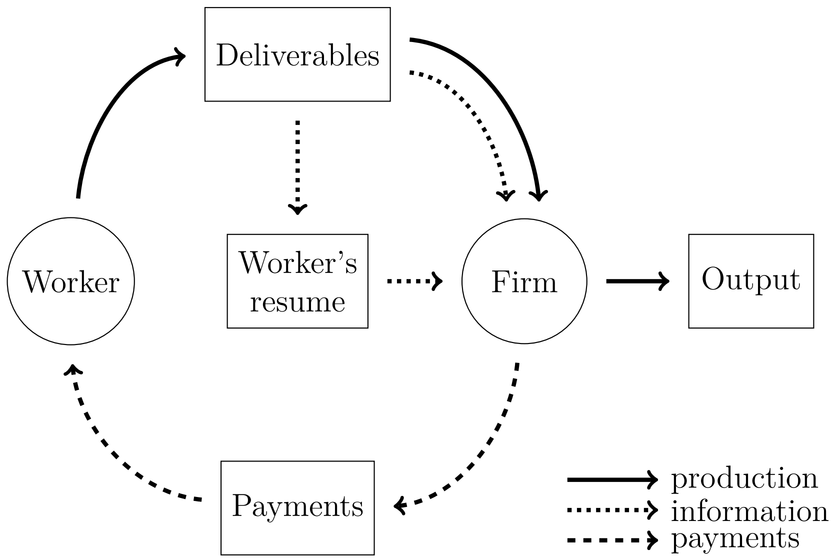


Figure 1: Flow of production, information, and payments

past) age of the workers at every moment they complete any amount of deliverables.

Firms, moreover, are able to infer the expected productivity given the signal of experience, either because they have hired many employees with the same experience and have seen how much output these employees have generated on average or because they know the economy-wide distribution of productivities and labor supply.[5] Finally, there is free entry and exit, so that salaries will be equal to the expected productivities of the workers, given their resumes.

Figure 1 represents diagrammatically the flows of production, payments, and information. Workers complete deliverables for the firms, and the firm sees the completion of the deliverables and the workers' resume. Each worker's resume keeps track of their deliverables completion history, adding up and weighing those the deliverables. Payments are based on what the firm observes, that is, resumes and the execution of the deliverable. Output and profits are realized, but the firm cannot individually assess the contribution of each unit of deliverable to the firm output.

2.3 Taxes

This model features what has been called "double adverse selection" (Stantcheva, 2014). Neither firms nor the government observe workers' types. We assume that firms

[5]The fact that firms observe hours but not individual output can also be thought of through the lens of team production (Alchian and Demsetz, 1972). Production needs to take place inside a firm that aggregates the work of multiple workers. For example, the firm production function could be $F=\prod_i^n \mathbb{1}(h_i > 0) \cdot \sum_i^n h_i \cdot v_i$, that is, production inside the firm is linear but it needs multiple (n) workers to be present. In the limit of a large number of workers (n), the firm would assess the productivity of workers only from their average productivity.

observe the workers' resumes, while the government observes histories of earnings $\tilde{y}$.[6] This assumption captures the idea that every year, tax payers send their tax returns to the the tax authority, and the tax authority keeps track of these tax returns. The government faces a budget constraint and can save and borrow at the same exogenous discount rates the workers face. Because the workers face a lifetime budget constraint, the timing of tax payments and transfers will not be pinned down by the model. Thus, we denote $R(\tilde{y})$ the lifetime post-tax earnings of the worker given the history of pre-tax earnings $\tilde{y}$. We will also denote $R(y)$ the lifetime post-tax earning of a worker given pre-tax lifetime income y, when taxes can be written as a function of the lifetime income y only, instead of full histories of earnings.

2.4 Equilibrium Definition

After describing the decision problem of workers and firms, as well as how the government sets taxes, this subsection defines a competitive equilibrium.

Definition. Equilibrium is described by workers choosing optimal consumption and labor supply flows, taking salaries and taxes as given, and anticipating the effect that their labor supply decisions have on their future salaries and taxes, as stated in Equation 1, while firms simultaneously set salaries according to the zero-profit Equation 2, presented below, paying each worker for their expected productivity conditional on their resumes. This zero profit condition is justified by competition among firms to enter the market and hire labor.[7]

For each marginal unit of labor, the firms' free entry condition is described by an Akerlof (1970) lemons condition,[8] where salaries are equal to productivity of the workers with the same resume:[9]

$$w(\mathtt{I}(\tilde{h}, a, \theta)) = \mathbb{E}[v(\theta)|\mathtt{I}(\tilde{h}, a, \theta)] \tag{2}$$

Equation 2 lies at the core of the model. It simply states that at any period workers with equivalent resumes will be paid equally, and that firms will on average break even. As in other signaling models, the condition above may not pin down uniquely the equlibrium. In particular, the condition does not restrict out-of-equilibrium beliefs.

Throughout this paper, we focus on cases where expected produvities conditional on the resume are determined in-equilibrium by Equation 2, that is, equilibria where for every achievable $\mathtt{I}$, there is a type θ, an age a, and an optimal choice $\tilde{h}$ so that $\mathtt{I} = \mathtt{I}(\tilde{h}, a, \theta)$.

[6]In Section 4, we discuss the extent that this assumption can be relaxed, and conditions under which optimal taxes take an even simpler form.

[7]See Attar et al. (2011) for a game theoretical foundation, in a static context.

[8]This condition perhaps would more precisely be named in this context, an Akerlof "peach condition," since under the assumption that those who are more willing to work are those who are more productive, there is advantageous selection instead of adverse selection.

[9]In this definition, it is assumed that $\mathtt{I}$ and $\tilde{h}$ are continuously distributed, so that the expectation is the same if it is conditioned on the workers who accept the contract.

In that sense, we will focus on separating equilibria, and ignore possible pooling equilibria where expectations may not be pinned down in-equilibrium. Additionally, when dealing with optimal taxation, we will make the usual assumption in the mechanism design literature that the planner can choose the equilibria if there are multiple.

3 Two Polar Cases

In this section, two polar cases illustrate a range of possibilities this simple setup can accommodate. These two benchmark cases are illustrative of the patterns of wages and the return to experience, as well as of the type of distortions that dynamic labor market signaling can generate.

In the first case, each worker's resume is summarized by the cumulative discounted sum of what the worker has produced or its "length," that is, $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a q(\tilde{a})\tilde{h}(\tilde{a})d\tilde{a}$. Payments per unit of the deliverable will be, for every worker, increasing with experience and almost at all times different than their marginal product of labor. In spite of salaries being higher at advanced stages of the career of the workers, there will be no intertemporal distortions. A dynamic rat race will generate high-powered incentives for workers to exert effort throughout their careers.

In the second case, each worker's resume is summarized by the "pace" of completion of deliverables: the ratio of the total amount of deliverables a worker has produced and the age of the worker, that is $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a \frac{\tilde{h}(\tilde{a})}{a} d\tilde{a}$. Payments per unit of the deliverable will be constant almost at every point, throughout the career of the workers, and equal to their marginal product of labor. In spite of pre-tax salaries being constant, corrective taxes will be necessary to avoid large intertemporal distortions, and excessively strong incentives to work hard at early stages of the career of workers. A dynamic rat rate will generate high-powered incentives to exert effort, but those incentives will be declining over time.

In Appendix B.1, we discuss more general properties of wages and the returns to experience that arrive at from arbitrary weighting functions, that is, when the resume is summarized by a signal of the form $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a \tilde{h}(\tilde{a})\phi(\tilde{a}, a)d\tilde{a}$, with $\phi(\tilde{a}, a) > 0$, bounded and continuous. We show that three properties hold generally: i) all workers start with an empty resume and earn the same initial salaries; ii) at the peak of the career of each worker, salaries are higher than productivities; iii) there is a positive signaling return to experience.

3.1 First Case: The Length of the Resume

The first benchmark case, or example of the informational structure we just presented, is one where the worker's resume is summarized by the cumulative discounted sum of what the worker has produced or its "length," that is, $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a q(\tilde{a})\tilde{h}(\tilde{a})d\tilde{a}$. This case has features that are closely analogous to Arrow (1962)'s learning-by-doing model, where productivity is a function of the cumulative use of a factor. Similarly here, firms' perceptions about worker's productivity are a function of the worker's cumulative labor supply.

This case is natural if most or all of the heterogeneity in labor supply is at the extensive margin, and if, for firms, it is hard to observe anything other than how many years or hours of experience a worker has. In fact, in many occupations, it is hard for firms to observe and verify much more than the start and end date of previous positions, or the number of hours someone has been working in the current position.

This case is also natural if firms believe that there is a lot of human capital accumulation, so that the model they may have in mind is one where there is on-the-job learning, in the form of the Arrow (1962) learning-by-doing model.[10] In that vein, firms could mistake the return to experience as human capital accumulation instead of changes in the composition of the pool of workers. That is, the firm could hire many workers with different lengths of resumes, and observe that the pool of workers with longer resumes is, on average, more productive. Without knowing why the pool is more productive, it may attribute that fact to human capital accumulation instead of signaling. Firms could think the change in the composition of workers who happen to achieve a higher degree of experience is a real return to human capital when it is not, and still post the same salaries and make zero profits. Workers as well may not know whether they will become more productive by working more, or whether they would just signal to employers that they are more productive. From the workers' perspective, to make their labor supply choices, it only matters how experience accumulation and labor supply decisions today will likely impact their future salaries.

Under the assumption that resumes are summarized by their "length," salaries satisfy a modified Akerlof condition, as stated in Lemma 1.

Lemma 1. *If the information of firms is defined as the "length-of-the-resume," that is,* $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a q(\tilde{a})\tilde{h}(\tilde{a})d\tilde{a}$*, then the Akerlof condition (Equation 2) is equivalent to:*

$$w(h) = \mathbb{E}[v(\theta)|h(\theta) \geq h] \tag{3}$$

Where $h(\theta)$ denotes the length of the resume of type θ at the end of their career, that is, $h(\theta) \equiv h_\theta(1) \equiv \int_0^1 q(\tilde{a})\tilde{h}_\theta(\tilde{a})d\tilde{a}$. Lemma 1 states that salaries as a function of the length of resume h are the average productivity of all the types who eventually reach a point in their careers where their resumes are longer than h. This is the case because at any period, for every type who eventually reaches a longer than h resume, there is someone,

[10] A more general version of the model that has both ingredients is presented and discussed in Section B.2.

potentially from a different generation, who has a resume of length h today.[11] However, if workers' labor supply decisions are heterogeneous only at the extensive margin (differing in how long they stay in the labor force) or if the deliverables were defined as working additional years, then the set of workers who would be pooled together would consist only of workers of the same cohort.[12]

Given Equation 3, which describes salaries as a function of the cumulative labor supplied, it is useful to note that lifetime income can be written quite simply. We can change variables and express lifetime income as integrating over increases in the length of the resume (dz), instead of as integrating over time, that is $y(h) \equiv y(h(1)) = \int_0^1 w(h(a))\tilde{h}(a)q(a)da = \int_0^h w(z)dz$. From that we can conclude that the additional amount a worker would receive in their lifetime from providing one extra unit of the deliverable (in present value) is the same at any point in the career of the worker. Moreover, this additional amount is equal to the payment for the last unit of the deliverable, $y'(h) = \mathbb{E}[v(\theta)|h(\theta) \geq h]$. These two observations simplify the analysis behind this dynamic model of career concerns, making it as simple as a static model.

In the economically-sensible case where it is less costly for the more productive people to supply the deliverables, salaries increase with experience. Workers at the beginning of their careers are willing to work more, relative to the myopic trade-off between current salaries and current effort, because they expect higher future salaries as an outcome of building up their experience, signaling to employers that they have higher productivities. The property that exerting effort increases future salaries, by affecting employers' perceptions of the ability of the worker, will be shared by a large range of informational structures, as described in more detail in Appendix B.1 and B.2.3.

Workers, when considering the timing of the completion of deliverables, have to choose between completing them earlier or later. If they complete deliverables earlier, they earn smaller direct payments from completing them at that time, but will enjoy larger benefits in the future from having built their reputations. Conversely, if they complete them later, they will earn larger direct payments from completing them, since their resumes will already be longer, but will enjoy smaller future benefits from building their reputation. It turns out that under the assumption that resumes are summarized by their length, these two forces exactly offset each other so the total discounted benefits from completing deliverables is constant throughout the careers of the workers.

At the beginning of their working life, for the first job they can get, all individuals face the same wage. Employers cannot distinguish between workers who have no experience at all. As workers advance in their careers, completing tasks and increasing their lifetime supply of labor, the length of the resume of hard-working individuals works to separate

[11]The same argument could be applied to any specification of $\mathtt{I}$ of the form $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a \phi(\tilde{a})\tilde{h}(\tilde{a})d\tilde{a}$.

[12]Given this structure of labor demand and supply, a natural question in this setup is whether there exists an equilibrium, as defined in Section 2.4. Proposition 11 in Appendix B.3, shows sufficient conditions for the existence of a separating equilibrium.

them from the other workers who execute fewer tasks and have shorter resumes. These more productive workers initiate their careers subsidizing the less productive, but at each new task they execute, some less productive workers are left behind with a shorter resume. For this reason, the remuneration that the more productive workers receive for the execution of tasks becomes progressively higher.

The size of the return to experience depends on how many people are being left behind by these more productive workers as they advance in their careers, and on how much more productive they are relative to those with shorter resumes. That is, the return to experience depends on the joint distribution of preferences and productivities.

Fortunately, there are simple and sensible assumptions on preferences and heterogeneity that allow us to derive equally simple expressions for the return to experience and the shape of the income distribution, as well as to analyze how they would change in response to information becoming more symmetric. Toward this goal, let's assume that preferences over lifetime labor supply and lifetime consumption are such that there is a constant elasticity of lifetime labor supply and those preferences are quasilinear in lifetime consumption.[13]

$$U(c,h,\theta) = c - \left(\frac{h}{b(\theta)}\right)^{1+\frac{1}{\epsilon}} \left(1+\frac{1}{\epsilon}\right)^{-1} \tag{4}$$

where $b(\theta) = \theta^{1-\delta}$,$v(\theta) = \theta^{\delta}$, $\theta \sim$ Pareto with shape parameter $\alpha > 1$, and $0 \leq \delta < 1$. Under this formulation $b(\theta)$ should be thought of as how many deliverables a worker can provide per unit of effort l, that is, $h = l \cdot b(\theta)$. $v(\theta)$ should be interpreted as how much output the worker generates per unit of the deliverable, that is, $y = h \cdot v(\theta)$. The total productivity as a function of effort, then, is just the product of $v(\theta)$ and $b(\theta)$, and it is equal to θ. In this example, δ governs the amount of information asymmetry: a higher δ means more heterogeneity comes from unobserved productivities ($v(\theta)$) instead of observable productivities ($b(\theta)$). Using the equilibrium definition and the convenient properties of Pareto distributions, we can guess and verify that salaries are also a power function, so wages and experience follow a log-linear relationship:

$$\log(w) = \gamma \cdot \log(h) + \kappa \tag{5}$$

where $\gamma = \frac{\delta}{1-\delta+\epsilon}$.[14] There is a constant proportional return to experience that is entirely driven by selection or employers learning about the types who are willing to take the jobs

[13]Note that given the result from Proposition 1, and assuming exogenous discount rates, it is without loss to specify preferences in terms of lifetime labor supply and lifetime consumption. This means that behind this preference specification there could be either heterogeneity only at the extensive margin (in which case, workers of different cohort would not be pooled together), or richer heterogeneity in preferences (in which case, workers of different cohorts potentially would be pooled together). This generality will be explored in our empirical estimation (Section 5), where we will focus on the extensive margin of labor supply.

[14]For the algebra behind this example, see Appendix A.1.

they are offering. This return to experience is larger when heterogeneity comes mostly from unobserved productivities, that is when δ is higher, as there is more heterogeneity to be screened out by experience. When the elasticity parameter ϵ is low, the return to experience is also larger: in this case, a higher experience is really indicating that the worker is more productive. When the elasticity is low, the difference between how a marginal increase in labor supply hurts the less productive relative to the more productive workers becomes larger. This equation illustrates that the observed pattern of wages as function of experience, which is often interpreted as coming from human capital acummulation, could alternatively be explained as coming from pure signaling (for the case of signaling with human capital accumulation see Appendix B.2.1). However, since the normative implications from signaling are very different than the normative implications of human capital accumulation, it is important to be able to write which moments of the data are informative about the possible distortions coming from signaling as opposed to human capital accumulation (as will be done in Section 4), as well as developing a strategy to disentangle human capital accumulation from signaling (as in Section 5).

Because higher types are more productive and are the workers who are willing to work longer, wages increase over time, and at the time of retirement, each worker would be facing a higher salary than their productivity. As noted earlier, whenever the resume is summarized by its "length," the lifetime benefits of increasing the lifetime labor supply do not depend on when the worker completes these extra units of the deliverable. Therefore, workers face high-powered incentives to work not only at the time of retirement, but at all moments in their careers. This implies that Pigouvian component of taxes as defined in Section 4 will be positive, and will correct for that distortion. This also will imply that dynamic job market signaling, in this case, will not introduce intertemporal distortions, and the optimal tax base will be the workers' lifetime earnings.

To summarize, whenever the public signal the firms can obtain from workers is the "length-of-their-resumes," payments per unit of the deliverable will be, for every worker, increasing with experience, and almost at all times different than their marginal product of labor. In spite of salaries being higher at advanced stages of the career of the workers, there will be no intertemporal distortions. A dynamic rat race will generate high-powered incentives for the workers to exert effort throughout their careers. Pigouvian taxes can be used to correct for these high-powered incentives as will be discussed in Section 4, taking lifetime income as an optimal tax base. In a simple parametric example, the return to experience will take a familiar log-linear form, and marginal Pigouvian taxes will be constant, implying that they can be implemented with history-independent annual taxes.

3.2 Second Case: The Pace of the Resume

This section presents the second benchmark case, where the resume is summarized by the pace of completion of deliverables: the ratio of the total amount of deliverables a worker has produced and the age of the worker, i.e. $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a \frac{\tilde{h}(\tilde{a})}{a} d\tilde{a}$. This case is particularly natural if all the heterogeneity in preferences are at the intensive margin, and employers are aware of the relationship between the heterogeneity in productivities and the willingness to supply deliverables. We will show that in this example, payments per unit of the deliverable will be constant throughout the career of the worker, and equal to their marginal product of labor. However, the corrective component of taxes will be positive and will correct for intertemporal distortions and the incentives to work excessively when young.

In this section, we assume that workers have additively separable preferences over time, and preferences are heterogenous only on the intensive margin. That is, every worker starts and ends their careers at the same age, but each of them may be willing to supply deliverables at different rates. In other words, we can think of all the heterogeneity as coming from how each worker evaluates different paces of work, and not from how long they would like to stay in the labor market. Under that assumption, any optimal allocation has each worker consuming and supplying labor at constant rates over their lifetimes. As we will see in more detail in Section 4, the government can implement those allocations using age-dependent and history-dependent taxes.

Moreover, whenever there is separation, more productive types will supply labor at higher rates, and therefore will have stronger resumes. Because they supply labor at constant rates, each worker will also reveal a constant "pace-of-the-resume," and each type, being associated uniquely with a certain "pace," will receive a different pre-tax salary, which will be equal to their marginal productivity. Therefore, for each worker, salaries will be equal to their marginal productivity almost at every point in their careers.[15]

The fact that pre-tax salaries are equal to the productivity of the workers does not mean, however, that there are no distortions that taxes should take care of. In fact, taxes should correct for potentially large intertemporal distortions that arise from the dynamic reputation building effects. If young workers increase their labor supply, they would receive not only larger payments now, but they would also improve their resumes, and increase the remuneration they would receive for the future completion of deliverables.

The size of the reputation building effects, and the return to experience depend on the joint distribution of productivities and willingness to provide the deliverables. Fortunately, as in the previous example from Section 3.1, there are simple parametric assumptions that allow us to derive simple expressions for the signaling return to experience and for the corrective component of taxes. In particular, for illustrative purposes, let us assume

[15] The exception being a zero measure moment when they have just entered the workforce and have empty resumes.

that preferences are described by Equation 6, below:

$$U = \int_0^1 e^{-\rho\tilde{a}} \left(\tilde{c}(\tilde{a}) - \frac{\tilde{h}(\tilde{a})^{1+\frac{1}{\epsilon}}}{1+\frac{1}{\epsilon}} b(\theta)^{-(1+\frac{1}{\epsilon})} \right) d\tilde{a}, \tag{6}$$

where $b(\theta)$ is the cost of a worker of type θ to work more and provide more deliverables per unity of time. We denote productivities by $v(\theta)$, and assume production is linear in the flow of labor supply. Productivities and preferences are parameterized so that $v(\theta) = \theta^{\delta}$, $b(\theta) = \theta^{1-\delta}$, with $0 < \delta < 1$. Discount rates are $q(a) = e^{-\rho a}$.

Any constrained optimal allocation has each workers supplying constant labor flows over time. Moreover, for concreteness, let's focus on the allocation where the redistributive component of taxes is set to zero, so that for each worker the marginal rate of substitution between consumption and leisure is equal to the marginal productivity; that is, the optimal flow of labor supply satisfies the following condition: $\tilde{h}(\theta)^{\frac{1}{\epsilon}} b(\theta)^{-(1+\frac{1}{\epsilon})} = v(\theta)$. Denoting a particular level of a resume pace as $\bar{h}$, using the condition above we can conclude that pre-tax salaries as a function of the pace would satisfy $w(\bar{h}) = \mathbb{E}[v(\theta) | \frac{\int_0^a \tilde{h}(\tilde{a}) d\tilde{a}}{a} = \bar{h}] = \bar{h}^{\frac{\delta}{1+\epsilon-\delta}}$. Notice that these are exactly the productivity of the workers for whom it is optimal to supply a certain pace $\bar{h}$. As in the previous case, wages still satisfy a log linear relationship, but in Equation 7, below the "pace" replaces the "length:"

$$\log(w) = \gamma \cdot \log\left(\bar{h}\right) \tag{7}$$

where $\gamma = \frac{\delta}{1-\delta+\epsilon}$. Salaries at every moment are equal to the marginal productivity of the workers. However, this does not imply that there should be no corrective taxes; in fact, the relevant labor wedge is not the static difference between productivities and current salaries, but the difference of productivities and the sum of current salaries and the increases in future lifetime earnings from reputation builiding effects. When a worker increases their labor supply at period a, their wages in future periods $a' > a$ increase since the future "pace" $\bar{h}$ is affected. In fact, the elasticity of their future salaries to current effort satisfies Equation 8, below. That is, the increase in future salary at time a' is equal to the ratio of γ and a', since one extra deliverable today contributes to a $1/a'$ higher "pace" at age a'.

$$\frac{d \log w(\bar{h}(a'))}{d \log \tilde{h}(a)} = \frac{\gamma}{a'} \tag{8}$$

Pigouvian taxes should exactly undo the positive effects of reputation building, as it is discussed in more detail in Section 4.2. Those Pigouvian taxes on earnings, in this example, will depend exclusively on age, and will be the highest at the beginning and gradually fall to zero as workers approach the end of their careers. They correct not for the difference between current payment and productivities (which in this example is zero!), but for the difference between the lifetime gains from a marginal increase in effort today and the productivities.

The dependence of the corrective taxes on age suggest that, without them, younger workers would exert a lot more effort and work longer hours. This prediction is in line with Landers et al. (1996) who show that, in law firms, hours decrease with tenure and associates work too many hours relative to their desires as well as too many hours relative to partners. This prediction, however, is at odds with the commonly found inverted U-shaped pattern for hours over the life-cycle: younger and less experienced workers work fewer hours, hours are mostly constant for workers between the ages of 25 to 55, and they quickly fall for older workers (Card, 1991; Kaplan, 2012). Importantly, both pieces of evidence could be alternatively explained by a combination of age-dependent preferences and age-dependent patterns of human capital accumulation and depreciation. However, the normative implications of human capital accumulation and signaling would be very distinct. For that reason, Section 5 will rely instead on exogenous changes in salaries and additional measures of productivity to disentangle the role of signaling from alternative explanations and, at the same time, quantify, from a normative point of view, the overall magnitude of the necessary Pigouvian corrections.

To summarize, whenever the public signal the firms can obtain from workers is the "pace-of-their-resumes," payments per unit of the deliverable will be, for every worker, constant throughout their lifetime, and almost at all times equal to their marginal product of labor. In spite of salaries being constant, there will be large intertemporal distortions. A dynamic rat race will generate high-powered incentives for the workers to exert effort throughout their careers, but those incentives will be stronger for younger workers and will decline as workers approach their retirement age. Pigouvian taxes can be used to correct for these high-powered incentives and intertemporal distortions as will be discussed in Section 4.

4 Taxation

In this section, we focus on the normative implications of the model and how dynamic signaling effects should affect optimal taxation. We first focus on the simpler case where the public signal that the firms see is defined as the "length-of-the-resume," or the cumulative discounted sum of what the worker has produced. Then, we extend the results to more general information structures.

As in Mirrlees (1971), the government does not observe the workers' types, and because firms also do not observe the types, this model features what has been called "double adverse selection" (Stantcheva, 2014). The general assumption we will make is that the government, instead, observes the history of earnings $\tilde{y}$, and the government will potentially uses that rich information to tax the workers.

In the first case, where resumes are summarized by their "length," we are going to place special focus on the case where the government uses a simple tax base, and conditions taxes only on lifetime income (y). This focus is motivated by Proposition 1 which extends Atkinson and Stiglitz (1976) uniform taxation result to show that in the "length-of-the-resume" case, it is still optimal to tax lifetime income even with "double adverse selection." Moreover, on the practical side, many taxes and transfers actually condition on lifetime income: most notably, the key determinant of US social security benefits is a measure of the average earnings over the lifetime of a worker.

Proposition 1. *If preferences are homogenous over the timing of consumption and labor supply flows and resumes are summarized by their "length"– that is, preferences take the form* $U(C(\tilde{c})), H(\tilde{h}), \theta)$ *and* $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a q(\tilde{a})\tilde{h}(\tilde{a})d\tilde{a}$ *–, then it is optimal to tax lifetime income, even with "double adverse selection."*

Proof. See Appendix A.2. □

In this proposition, $C(\tilde{c})$ and $H(\tilde{h})$ are common aggregators of flows of consumption and labor supply, respectively. The assumption on preferences states that household preferences are homogenous over the timing of consumption and labor supply decisions; that is, preferences can be written as a function of common indexes (C, H) that aggregate the flows of consumption and labor supply, and these indexes are the same across households. Under this assumption, it is optimal to use only lifetime income as the tax base. The proposition says that when dynamic job market signaling does not introduce intertemporal distortions, and when preferences over timing of consumption and labor flows are homogeneous across households, then there are no reasons to introduce further distortions on intertemporal decisions.

Moreover, the content of this proposition is general in an important way. It essentially says that, when it is possible, post-tax lifetime earnings should be a function of lifetime labor supply. In the simple case where resumes are summarized by their "length," pre-tax lifetime earnings are a function of lifetime labor supply, and thus taxes on lifetime income are enough to guarantee that post-tax lifetime earnings are a function of lifetime labor supply. But away from this benchmark case, when pre-tax lifetime earnings are not necessarily a function of lifetime labor supply (for example, if the signal the firm sees about a worker is a different function of the flows of labor supply instead of the discounted sum of these flows, as in Section 4.2), the optimal policy would entail taxes meant to undo these intertemporal distortions. By doing that, the optimal policy would effectively make post-tax lifetime earnings a function of discounted lifetime labor supply.

Besides considering what happens when we relax the assumption that resumes are summarized by their "length," it is worth considering what happens when the homogeneity assumption is relaxed. In this case, even absent labor market distortions, the planner when thinking of redistributing income would want to use the intertemporal margin as a way of screening workers. Thus, the redistributive component of taxes would not have a simple base as lifetime income, even though the existing labor market distortions could be corrected using lifetime income taxes. When we simultaneously relax both assumptions, that resumes are summarized by their "length," and that preferences are homogenous over the timing of labor consumption and labor, then both the redistributive component of taxes and the corrective component of taxes would not have a simple tax base. This more general case is discussed in Appendix B.2.3.

4.1 Optimal Taxation Formulas

In this section, we derive necessary conditions for optimal taxes in terms of sufficient statistics, as in Saez (2001). We assume that the government maximizes a welfarist functional of worker utilities, $W(V(R;\theta),\theta)$, where R is the lifetime retention function function that maps pre-tax lifetime income (y) to post-tax lifetime income ($R(y)$).[16] The government solves the maximization problem:

$$\underset{R(y)}{max}\ \mathbb{E}[W(V(R;\theta),\theta)]\ s.t.\ \mathbb{E}[y(h(R;\theta),R)-R(y(h(R;\theta))]\geq 0 \tag{9}$$

subject, also, to the constraint that pre-tax wages are determined by the differential equation $y'(h)=\mathbb{E}[v(\theta)|h(R;\theta)\geq h]$ with initial condition $y(0)=0$.

Solving directly for $R(y)$ is complicated; even absent income effects, changing taxes around an income level y affects the incentives of workers with lifetime earnings around y, and by selection effects changes the joint distribution of resumes and productivities. Firms, recognizing that the pool of workers is different, would update the salaries of the affected workers. Salary updates causes another round of changes in the incentives of workers, which causes another round of changes in the pool of workers, and so on. Thus the original change in marginal taxes cause cascading effects. The change in the joint distribution of resumes and productivies is even more intricate when there are income effects: then the change in taxes and salaries around y affects not only the incentives of the workers at that income level, but the incentives of everyone above it. That would result in further salary updates at higher income levels, and generate new rounds of both compensated and income effects.[17] However, the following proposition allows us to simplify the problem by allowing the planner to keep salaries fixed when taxes change. In other words, we can frame the problem as the planner solving directly for post-tax salaries, effectively ignoring how pre-tax salaries are set.

[16]This formulation is quite general, and can be converted to Pareto efficiency tests as in Werning (2007); Hosseini and Shourideh (2019) and Bierbrauer et al. (2020), by picking linear functionals of the form $W(V(R;\theta),\theta)=\lambda(\theta)\cdot V(R;\theta)$, making it the dual of a revenue maximization problem subject to a minimum utility requirement for each type.

[17]This idea suggests a way to to test empirically whether the mechanism we highlight is at play. Indeed, in Section 5.2.2, we leverage the effects of changes in taxes on salaries as a source of identification for the labor wedges.

Lemma 2. *Without loss, we can solve directly for $\tilde{R}(h) = R(y(h))$, and then find $y(h)$, and $R(y)$. That is, the planner can solve the simpler problem:*

$$\max_{\tilde{R}} \mathbb{E}[W(V(\tilde{R};\theta),\theta)] \ s.t. \ \mathbb{E}[v(\theta)h(\tilde{R};\theta) - \tilde{R}(h(\tilde{R};\theta))] \geq 0 \tag{10}$$

Proof. See Appendix A.3. □

Lemma 2 applies more generally to other models of labor market frictions provided that firms make zero profits,[18] and, given any allocation, $y(h)$ is well defined and invertible. For example, there could be other production externalities, imperfect competition generating compressed wages, or information asymmetries and competitive or monopsonistic screening. This feature allows us to conceptually separate the relevant externalities coming from the information frictions that the planner would like to correct from transfers and innocuous price adjustments.[19] Importantly, the same idea will be applied in Section 4.2, to more general information structures, allowing us to separate issues of efficiency from issues of redistribution quite generally.

Proposition 2. *(For single-dimensional θ) If a tax schedule is optimal, then it satisfies the following optimal tax formula:*

$$\left(\frac{\chi(y) - r(y)}{r(y)}\right)\epsilon_r^c(y)g(y)y = \int_y^\infty \left(1 - \lambda(\tilde{y})\right)g(\tilde{y})d\tilde{y} + \int_y^\infty \left(\frac{\chi(\tilde{y}) - r(\tilde{y})}{r(\tilde{y})}\right)\eta_I(\tilde{y})g(\tilde{y})d\tilde{y} \tag{11}$$

where $r(y)$ denotes marginal retention; $\epsilon_r^c(y) = \frac{dy}{dr_y}\frac{r(y)}{y}$ are the local compensated elasticities; $g(y)$ denotes the density of y; $\lambda(y) \equiv \frac{W_V U_c}{\mu}$ is the marginal value the planner places on transfer to a worker earning y; $\eta_I(y) = \frac{dy}{dI}r(y)$ is the income elasticity; $\chi(y) \equiv \frac{v(y)}{y'(h(y))}$, and $v(y)$ is the productivity of the worker with lifetime income y.

Proof. See Appendix A.4. □

This equation is almost the same as the standard first-order condition that appears in Saez (2001) and Werning (2007), with one additional ingredient: $\chi(y)$. The equation says that if a tax schedule is optimal, then three sorts of effects should balance each other: compensated effects, mechanical and welfare effects, and income effects. When a planner considers increasing marginal retention over a small region (holding post-tax salaries fixed everywhere else), there are compensated effects (left-hand side of the equation) coming from the fact that people would work more. These effects are proportional to the densities and compensated elasticities, and they ease the feasibility constraint not by $1 - r(y)$ as in the standard Mirrleesian model, but by $\chi(y) - r(y)$, that is, a worker who earns an extra

[18] Or, more generally, there is full taxation of profits, or, still, profits are uniformly shared between workers, as in Scheuer and Werning (2017).

[19] Innocuous in the sense of the first and second welfare theorems, where changes in fundamentals could result in changes in relative prices without generating any inefficiencies.

dollar in their lifetime generates $\chi(y)$ units of output per dollar earned (and retains $r(y)$). The mechanical and welfare effects (first term on the right-hand side of the equation) are the same as before: the planner is giving one dollar to those who earn at least y, and this has mechanical costs of one dollar per person as well as welfare effects that are weighted by the marginal value of a dollar that the planner attributes to a transfer to each of these people. Finally, there are income effects that affect everyone who earns at least y. These people are induced to work less (if income effects are negative), and, as in the compensated effects, they damage the feasibility constraint by $\chi(y) - r(y)$.

The same equation can alternatively be written in two blocks. The first block (Equation 12) defines retention as the product of two components, the Mirrleesian component, and a Pigouvian component that corrects for the production externality. The second block (Equation 13) translates the redistributive motive, and it is a standard Mirrleesian formula:[20]

$$r(y) = r_m(y) \cdot \chi(y) \tag{12}$$

$$\left(\frac{1 - r_m(y)}{r_m(y)}\right)\epsilon_r^c(y)g(y)y = \int_y^\infty \Big(1 - \lambda(\tilde{y})\Big)g(\tilde{y})d\tilde{y} + \int_y^\infty \left(\frac{1 - r_m(\tilde{y})}{r_m(\tilde{y})}\right)\eta_I(\tilde{y})g(\tilde{y})d\tilde{y} \tag{13}$$

This decomposition allows us to understand what are the externalities that career concerns make workers impose on each other. The key insight is that we can think of the difference between payments and marginal products for each worker, holding remuneration schedules fixed, instead of thinking how the individual actions of these workers affect the remuneration schedule of other workers. When increasing their lifetime labor supply so that they get paid one extra dollar, they are contributing to the economy not one dollar but $\chi(y)$ dollars. In other words, it is as if they would produce one dollar but generate negative production externalities of the size of $1 - \chi(y)$ dollars.

It is interesting to compare the following economies in terms of their utility possiblities frontiers: (a) the economy without any information asymmetries, (b) the economy with information asymmetries only between the government and the workers, as in the standard Mirrlees model, and (c) the economy with information asymmetries between the firms, workers and government. In (b), when there are no information asymmetries in the labor markets, the planner could set taxes to zero, and then by the first welfare theorem, we know this is a first best allocation. Hence, there is a common point in the first and second best utility possibilities frontiers, i.e., in the utility possibilities frontiers of (a) and (b). Now, what is perhaps surprising is that there is a common point in the utility possibilities frontiers of (a) and (c), that is, between the third best utility possibilities frontier and the first best utility possibilities frontier, which is achieved by setting the Mirrleesian component of taxes to zero, so that $r(y) = \chi(y)$, as shown in Proposition 12, in Appendix B.4. This further justifies the decomposition above as one between a Mirrleesian and a

[20]As it appears, for example, in Scheuer and Werning (2017) and Saez (2001).

Pigouvian component.

Employers, when facing workers with the same resume, do not know which workers are supplying their last unit of labor and will not extend their resumes further. By the assumption that those who are more willing to extend their resumes are those who are more productive in the unobservable dimension, the workers who are retiring have the smallest productivity among those with the same resume. Therefore, we have that $\chi(y) \leq 1$ workers supplying their last unit of labor are paid more than their marginal products. From the tax formula above, holding estimates of elasticities and densities of the income distribution constant, we can see that taking into account these career concerns unambiguously pushes toward higher marginal taxes at every income level.

These optimal taxes do not depend only on labor wedges but also on elasticities and the shape of the income distribution. From the point of view of optimal taxation formulas, similarly to results in Scheuer and Werning (2017), common elasticity estimates here are biased downwards. The reason for this bias are is the following general equilibrium logic: increases in marginal retention induce the marginal types – who are the less productive types – to work more, and therefore reduce pre-tax salaries, making the effective change in post-tax wages smaller. Thus, the estimated elasticities of taxable income are lower in magnitude than the elasticities that enter optimal taxation formulas, which keep pre-tax salaries fixed.[21] Moreover, the elasticities in the formulas are long-term elasticities of lifetime income. These may differ and may be higher than commonly estimated short term elasticities of annual income, as discussed in Imai and Keane (2004); Saez et al. (2012) and Kurnaz et al. (2023).

The marginal Pigouvian component of taxes can be thought of as guaranteeing that workers at the margin are paid for their marginal product, i.e. post- Pigouvian tax payments for an additional increase in lifetime labor supply marginal unit of labor ($r(h)$) should be such that $r(h) = v(\theta)$. We can translate this condition as a restriction on marginal retention on earnings ($r(y)$), which states that $\chi(y) = \frac{v(\theta(y))}{\mathbb{E}[v(\theta)|y(\theta)\geq y]}$. Under the parametric assumptions from Section 3.1, where productivities are Pareto distributed, Pigouvian taxes are given by $\tau_p(y) = 1 - \chi(y) = \frac{\delta}{\alpha}$. Pigouvian taxes are increasing in the amount of information asymmetry δ, and are constant as a function of lifetime income. Thus, in the parametric example, the Pigouvian corrections can be implemented with linear taxes. This makes the implementation particularly simple: the Pigouvian corrections can be implemented with history-independent annual income taxes.[22]

It is useful to compare the results from this case to the Miyazaki-Wilson-Spence (MWS) model of labor markets, also analyzed in Stantcheva (2014). We have shown that the "length-of-the-resume" model admits a static representation. That representation allows

[21]For more details see Sztutman (2023), Chapter 1, Appendix 1.9.7.

[22]Linearity is one property that guarantees that lifetime income taxes can be implemented with history-independent annual taxes. Besides linearity, whenever there is heterogeneity only at the extensive margin, lifetime income taxes can be implemented with history-independent non-linear annual taxes.

us to contrast it to the static MWS model. In each case, salaries are a different function of effort. In our case, the total income a worker receives when supplying h is equal to $\int_0^h \mathbb{E}[v(\theta)|h(\theta) \le \tilde{h}]d\tilde{h}$. In their case, that income would instead be $h \cdot \mathbb{E}[v(\theta)|h(\theta) = h] = \int_0^h \mathbb{E}[v(\theta)|h(\theta) = h]d\tilde{h}$. This difference reflects the fact that, in their model, when workers decide to increase their labor supply, they enjoy higher payments for the inframarginal units of labor, which are then perceived to be supplied by a more productive worker. In constrast, here workers gradually distinguish themselves from their less productive peers, and increasing labor supply does not affect the payments for the inframarginal units of labor. On the other hand, from a dynamic perspective, the MWS model can be thought of as if regardless of where the workers are on their career trajectory, employers would pay workers based on the final resume they will achieve instead of the resume they currently have. Both models generate high-powered incentives, but the resume model generates implicit labor market cross-subsidies between workers that are absent in the MWS model.

These formulas can also be read as Pareto efficiency tests, as in Werning (2007). If a tax schedule is Pareto efficient, then there are weights $\lambda(y) \ge 0$ such that, given the current tax rates, the income distribution, and the estimated elasticities and labor wedges $\chi(y)$, the formula above holds. Relative to the standard Pareto efficiency test, the inclusion of $\chi(y)$, holding the other estimated statistics fixed, makes the formula easier to be satisfied; that is, higher marginal tax rates can be rationalized.

4.2 Taxes Under Richer Signal Structures

So far in this section, we have assumed that firms summarize a resume by looking at its length, which we have defined as the discounted cumulative sum of deliverables the worker has completed. Now, we relax this assumption and consider a richer set of signal structures under the same framework. In particular, we will consider signals that take the form $\mathtt{I}(\tilde{h}, a) = \int_0^a \phi(\tilde{a}, a)\tilde{h}(\tilde{a})d\tilde{a}$, with $\phi(\tilde{a}, a) > 0$, continuous in a and $\tilde{a}$.[23] The general idea behind this formulation is that the signal the firm sees can be thought of as the "strength of the resume." This measure is an imperfect signal of the past history of deliverables of the worker, but completing more deliverables always makes the resume stronger. General positive properties of this formulation are discussed in Appendix B.1.[24]

This formulation can capture different possibilities, including the previously discussed formulations of the "length-of-the-resume" and the "pace-of-the-resume" as special cases. Moreover, it can capture in reduced form a variety of possibilities: firms may mistake

[23]This assumption guarantees that expectations of productivities are well defined, and further it will be shown that those expectations are increasing in the completion of tasks $\tilde{h}$. But, more generally, we could consider any informational structure with these properties. Later in this section, the full history of labor supply decisions will be assumed to be observed, and there will be richer heterogeneity in preferences to guarantee that those expectations are well-defined.

[24]Appendix B.2.3 discusses the yet more general case where firms observe the full history of deliverables and there is richer heterogeneity in the type space.

human capital for signaling, firms may discount past experiences as they become harder to verify, or may discount recent experiences as new information may arrive with delays. as long as, when put together, they can be summarized by the idea that firms would evaluate the experience through the lens of an index of the form $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a \phi(\tilde{a}, a)\tilde{h}(\tilde{a})d\tilde{a}$. This formulation bypasses the need to introduce explicitly all those elements, and to postulate explicit stochastic processes for shocks for each of those elements and considerations. Not introducing those shocks directly makes the analysis of optimal tax systems tractable, and avoids technical issues arriving from multidimensional screening problems, as well as from failures of the homogeneity assumption on preferences over the timing of consumption and labor supply flows.[25]

Despite this apparent complexity, optimal taxation formulas will take a simple structure. We will follow the same logic from Proposition 2, and assume preferences are homogeneous over the timing of labor supply and consumption, and that more productive workers are more willing to provide the deliverables.[26] Under these assumptions, it will be shown that, as in the previous section, taxes can be described as the composition of i) corrective taxes that guarantee that the workers benefits from each marginal increase in labor supply are equal to their contribution to the output of the firm, and ii) redistributive taxes that are described by the same Mirrleesian optimal taxation formulas.

We will first set aside the issue of implementation, and characterize the set of optimal incentive compatible allocations. Those allocations need to satisfy two properties, more formally stated in Lemma 3, in Appendix A.5. These properties state that, ii) intertemporally, labor supply decisions will not be distorted, and ii) an analogous equation to the standard Mirrleesian formula should hold in terms of the lifetime labor supply. Moreover, because of the homogeneity assumptions on preferences, Lemma 3 and Proposition 3 imply that correcting for intertemporal distortions can generate Pareto improvements, as demonstrated in Appendix A.5.

Then, toward the goal of implementing these optimal allocations with a system of taxes and transfers, we establish that, quite generally, there is a positive return to experience – formally stated in Lemma 4 (in Appendix A.5). In other words, exerting more effort is a way for workers to signal to employers they are more productive, and thus, exerting more effort will impact future salaries positively. Intuitively, the signals considered in this section are increasing in the effort decisions, and under the assumption that those who are willing to exert more effort are those who are more productive, employers can infer that higher signals translate into higher expected productivities. Besides being of interest in itself, this result is used to show that the planner can infer labor supply decisions from

[25]One may worry that those weights should be endogenous to the tax system. However, as will be shown, the same optimal taxation formulas will hold even if those weights were to depend on the tax system.

[26]Which in this case follows from additionally assuming that $\frac{d^2 H(\tilde{h})}{d\tilde{h}(a')d\tilde{h}(a)} < 0$.

earnings histories, as formally stated in Lemma 5 (in Appendix A.5).

The result in Lemma 5 allows us to decentralize the incentive compatible and efficient allocations, analogous to the way that Lemma 2 was used to derive the optimal lifetime income taxation formulas from Proposition 2. That is, we can think of the planner as solving for the optimal allocation as in Lemma 3, or equivalently, solving for post-tax wages $\tilde{R}(\tilde{h})$, and then implementing this post-tax retention function with history-dependent earnings taxes.

Finally, we summarize these taxation results in Proposition 3, below, which states that the tax system should be such that i) history and age-dependent taxes (R_p) should be used to correct for labor wedges, and ii) after correcting for these distortions, lifetime income redistributive taxes should be imposed on top of these taxes, according to standard redistributive formulas.

Proposition 3. *If $R(\tilde{y})$ is optimal then there are R_m, R_p with $R(\tilde{y}) = R_m(R_p(\tilde{y}))$, such that R_m and R_p satisfy the following conditions:*

$$v(\tilde{h})q(a) = \frac{dR_p(\tilde{y})}{d\tilde{y}(a)}w(\tilde{h}^a) + \int_a^1 \frac{dR_p(\tilde{y})}{d\tilde{y}(\tilde{a})}\frac{dw(\tilde{h}^{\tilde{a}})}{d\tilde{h}(a)}\tilde{h}(\tilde{a})d\tilde{a}, \tag{14}$$

$$\left(\frac{1-r_m(R_p)}{r_m(R_p)}\right)g(R_p)R_p\epsilon^c_{\bar{r}}(R_p) = \int_{R_p}^{\infty} g(\tilde{R}_p)\Big(1-\lambda(\tilde{R}_p)\Big)d\tilde{R}_p + \int_{R_p}^{\infty}\left(\frac{1-r_m(\tilde{R}_p)}{r_m(\tilde{R}_p)}\right)g(\tilde{R}_p)\eta_I(\tilde{R}_p)d\tilde{R}_p, \tag{15}$$

where $r_m = R_m'(R_p)$, and $v(\tilde{h})$ is the marginal productivity of the type that supplies labor flows $\tilde{h}$, and where for ease of notation the dependence on $\tilde{y}$ is omitted. That is, the formula should be read as a function of earnings flows $\tilde{y}$, through the inverse operator $\tilde{h}(\tilde{y})$.

Notice that if there is a tax system in place that already corrects for intertemporal distortions R_p, then Proposition 3 tells us that we are back to the simpler case covered in Proposition 1. Moreover, as a corollary of Proposition 3, we know that if the information structure does not introduce intertemporal distortions, then the simpler optimal taxation formulas from Equation 11 hold.

Corollary 1. *If the information structure is such that no intertemporal distortions are generated, then if taxes are optimal they satisfy Equation 11.*

Notice that correcting for intertemporal distortions can be a significantly more complicated endeavor: taxes can be history-dependent, and will depend on how much a change in labor supply today translates into higher lifetime earnings, not only through its impact on current earnings, but additionally through its indirect impact on future salaries.However, in the "pace-of-the-resume" example from Section 3.2, Pigouvian taxes become surprisingly simple, can be solved for explicitly, and depend only on age, as stated in Corollary 2.

Corollary 2. *If resumes are summarized by their "pace," under the parametric assumptions of Section 3.2, then* $\frac{dR_p(\tilde{y})}{d\tilde{y}(a)} = a^{\gamma} e^{-\rho a}$*, where* $\gamma = \frac{\delta}{1+\epsilon-\delta}$*.*

Proof. See Appendix A.5. □

The Pigouvian taxes in the "pace-of-resume" case correct not for the difference between marginal productivities and salaries (which in fact are equal to zero) but for the dynamic reputational effects that generate high-powered incentives for young workers. To accomplish that, Pigouvian taxes need to be high for young workers and slowly decrease to zero as workers approach the end of their careers.

The normative results in this section can be compared to the results in Holmström (1999). In both models, workers enter a dynamic rat race. When the resumes in our signaling-by-doing model are summarized by their length, there are no intertemporal distortions, as the special case in Holmström (1999), where there is no discounting and productivities follow a random walk process. When resumes are summarized by their pace, reputation concerns induce workers to exert relatively more effort earlier in their careers, and progressively exert less effort as their reputations are consolidated. This is also the case in Holmström (1999). However, the distortions in Holmström (1999), in contrast, can result in workers exerting too little effort towards the end of their careers, as firms are not allowed to pay for perfomance and the workers' reputations are consolidated. This possibility does not arise here as firms can pay the workers directly for their observable effort $\tilde{h}$ (i.e. the "deliverables").

4.3 Changes in Information Processing Technologies

As new technologies and richer datasets begin to affect the workplace (Chalfin et al., 2016; Autor, 2019; Acemoglu et al., 2020; Bales and Stone, 2020), an important question is the impact of these technological changes on the workers' incentives to exert effort, on the distribution of income, and on the tradeoffs the government faces when setting taxes. Those changes affect how workers get paid and how resumes are read, arguably making information imperfections less pronounced. On the other hand, the rising automation of routine tasks (Autor et al., 2003), may have contributed to changes in the task composition of jobs and to an increasing prevalence of non-routine cognitive tasks. As a result, it may have become harder to monitor and assess the productivity of workers, and information imperfections in the labor markets may have become more salient. In this section, we look at the consequences of changes in fundamentals that affect information asymmetries in labor markets.

We say that information asymmetry problems in labor markets decrease if deliverables become a better measure of product, affecting how firms pay workers and how firms read resumes. More precisely, we adopt the following definition.

Definition. Let preferences be $U(c,h,\theta) = \tilde{U}\Big(c, h/b(\theta)\Big)$. If new productivities and tastes are such that $v^n(\theta) = v(\theta)/\Delta(\theta)$ and $b^n(\theta) = b(\theta)\cdot\Delta(\theta)$, $\Delta > 0$ and $\Delta' > 0$, order preserving (meaning that if $\theta > \theta'$, then $v(\theta) > v(\theta')$, $v^n(\theta) > v^n(\theta')$, $MRS^{\theta}_{c,h} > MRS^{\theta'}_{c,h}$, and $MRS^{n,\theta}_{c,h} > MRS^{n,\theta'}_{c,h}$, where $MRS^{\theta}_{c,h} \equiv -\frac{U_h(c,h,\theta)}{U_c(c,h,\theta)}$), then information asymmetries in labor markets decrease.[27]

Under this definition, decreasing information asymmetries keeps the first-best utility possibilities frontier unchanged. It also increases the ratio $v(\theta)/\mathbb{E}[v(\tilde{\theta})|\tilde{\theta} > \theta]$ for every type of worker, and preserves the relative ranking of types in terms of productivities and marginal rates of substitution.

A perhaps surprising result is that information asymmetries in labor markets increase welfare in this setup, as stated in Proposition 4. This result implies that those frictions are, in a particular way, good for redistribution, and attenuate inequality. That is, if the way in which the government evaluates inequality is expressed in how it sets marginal tax rates, then an increase in information asymmetries in labor markets increases welfare, or, in other words, improves the distribution of outcomes from the point of view of a redistributive planner. Information asymmetries help with redistribution, because they make it harder for high productivity workers to separate themselves from low productivity workers. To achieve higher salaries, high productivity workers have to show they are productive by supplying labor more intensively than low productivity workers are willing to do. Moreover, because under the assumptions of Lemma 2 (or the more general Lemma 5), the optimal post-tax wages do not depend on the specifics of the model of asymmetric information, the impact on welfare is the same across different models, provided that taxes are set optimally and following the same preferences for redistribution.

In other words, this result does not depend on the particular assumption we imposed on the information structure that determines how salaries are set by firms. It relies on the planner being able to solve for the allocation directly, which is possible whenever there is a one-to-one mapping between earnings and labor supply decisions. This property is satisfied by different information structures as shown in Section 4.2. Further, the result also holds across different models of career concerns, provided they satisfy the invertibility conditions behind Lemma 2 or the more general Lemma 5, which allow for labor supply decisions of workers to be perfectly inferred from their history of earnings. In fact, this result is a generalization of Stantcheva (2014) Proposition 13, which compares welfare in an economy under a Miyazaki-Wilson-Spence (MWS) model of the labor market relative to a Mirrleesian economy.[28]

[27] Notice that under the assumptions behind Proposition 1, any optimal allocation makes post-tax wages a function of lifetime labor supply. Thus, in this case, it is without loss to specify preferences as a function of lifetime labor supply and lifetime consumption.

[28] Another similar setup where these conditions are satisfied is the Azevedo and Gottlieb (2017) model of competitive screening.

Proposition 4. *If the original tax schedule is optimal and the planner has redistributive preferences, workers' preferences are convex over leisure and consumption, and leisure is a normal good, then decreasing information asymmetries in labor markets decreases welfare.*

Proof. See Appendix A.6. □

The idea behind this result is that if the planner has redistributive tastes, then it sets positive marginal redistributive taxes to transfer resources from higher types to lower types, and the incentive compatibility constraints bind downwards (Seade, 1982; Werning, 2000). Whenever the degree of informational asymmetry in labor markets decreases, these downward incentive compatibility constraints become tighter, as it becomes less costly for the higher types to imitate the lower types. With less information asymmetry in labor markets, the high productivity workers can use more of their previously unobserved productivities to imitate the deliverable production of the lower productivity workers.

The impact on taxes is more subtle and it is discussed in more detail in Appendix B.5. When resumes are defined as the cumulative discounted sum of deliverables, the corrective component unambiguously falls for each type, because $\chi(\theta) = \frac{v(\theta)}{\mathbb{E}[v(\tilde{\theta})|\tilde{\theta} \geq \theta]}$ increases proportionally more in the numerator relative to the denominator. In that vein, we can think of a decrease in information asymmetries in labor markets as a force toward lower taxes. However, the impact on the Mirrlesian component is in general ambiguous. Under the parametric assumptions of Section 3.1, the effects of the Pigouvian component dominate countervailing effects on the Mirrlesian component in such a way that, holding fixed the marginal value of transfers λ, optimal taxes increase as information asymmetries increase.

5 Empirical Evidence

The previous section has offered an understanding of the tradeoffs a government faces between incentives and redistribution through the lens of simple sufficient statistics.

Surprisingly, in the first benchmark case from Section 3.1, where workers' resumes are summarized by the cumulative discounted sum of deliverables they have supplied so far in their careers, for the government to correct for the dynamic rat race distortions it is not necessary to keep track of the ratio of marginal productivities over salaries throughout the lifetime of the worker. Instead, it is enough to look at the ratio of marginal productivities over salaries for the last unit of labor that workers supply, that is, around the time of their retirement. This section estimates this ratio, and provides evidence that the estimated labor wedges are driven by asymmetric information in labor markets.

Moreover, in the second benchmark case from Section 3.2, where resumes are summarized by the pace at which workers have been supplying deliverables, it was shown that there can be larger intertemporal distortions. In particular, there are high-powered incentives for workers to exert effort, but the high-powered incentives decline over the career of the worker and reach the lowest level around the time workers supply their last unit of labor. In that sense, we can think of the results in this section as providing a suggestive number for the overall strength of the rat race, which in practice may be higher at earlier stages of the career of workers, and may additionally generate intertemporal distortions.

More generally, as discussed in Section 4.2, we would like to estimate the difference in marginal productivities and the sum of current salaries and all future earnings increases that result from supplying one unit of labor, at each point in the career of the workers. While estimating marginal productivities over salaries is a difficult challenge, estimating the latter is even harder. Direct attempts at estimating this richer set of statistics are left for future work.

5.1 Data

To evaluate the magnitude of the signaling return to experience and the Pigouvian component of taxes we use Health and Retirement Study data. The survey is a representative sample of the US population older than 50 and a biannual panel covering the period from 1992-2018. It follows about 20,000 individuals and it is rich on covariates, including job histories, hours worked, education, cognition, measures of lifetime income, and geographic, industry, and occupation variables.

There were several state and federal tax reforms over the period, which we are going to explore as a source of exogenous variation in wages.[29] To explore that variation, we construct simulated changes in marginal rates at original income levels by assigning a wide range of income, consumption and demographic variables from HRS to inputs in the NBER tax simulator (Feenberg and Coutts, 1993), extending the procedure developed by Pantoja et al. (2018).[30]

5.2 Empirical Strategy and Results

Our approach builds on the literature that has quantified the degree of adverse selection in markets described by simple Akerlof (1970) lemons conditions, such as Einav et al. (2010). In the context of health insurance markets they analyze, the key argument is that with a source of exogenous variation in prices, one can non-parametrically trace the shape of the cost curve for the insurance contract by looking at the average cost as a function of prices. Translated into our context, with a source of exogenous variation in wages (for a specific labor contract), one can non-parametrically trace the effect of selection by looking at the average productivities of workers who accept the contract as a function of wages.

[29] Appendix C.3 presents these in more detail.

[30] A more detailed description of the mapping between the variables is presented in Appendix C.2.

As the source of exogenous variation in post-tax wages, we will leverage tax reforms at the state and federal level. To parsimoniously separate different labor markets, we will pool together workers who, at a particular date, receive the same hourly wages, treating hourly wages as a proxy for their resumes. Finally, to infer expected productivities, two complementary approaches will be adopted. The first takes advantage of the rich set of covariates available from the Health and Retirement Study and looks at cognitive measures as proxies for productivities. The second assumes that labor markets are competitive and therefore leverages the observation that wages would be equal to the average productivity of workers with the same resume, as in the model presented in this paper. The first, presents evidence that selection, and thus asymmetric information, are behind changes in the pool of active workers. The second approach, while less robust, has the key advantage of allowing us to assign a number to a key quantity of interest: the labor wedges and the associated corrective component of taxes.

With that exercise, we will focus on the extensive margin of labor supply instead of the intensive margin. That focus is motivated by the observation that changes in the pool of workers with a certain resume driven by extensive margin responses are much more salient than changes driven by intensive margin responses. In the first case, marginal workers may either remain or entirely drop out of the labor force. In the second case, they would remain in the workforce but generate slightly weaker or stronger resumes. In the short run, these changes may be hardly noticeble to employers. Indeed, when extracting labor wedges from salary changes we will be assuming that changes in the composition of workers with a given resume coming form intensive margin effects are neglible relative to the size of extensive margin effects.

5.2.1 Cognitive Measures

In order to provide evidence that the pool of workers is changing in response to after-tax salary changes, we look at cognitive measures collected by the Health and Retirement Study. Conditional on initial hourly wages, after tax decreases, the working population's average mental status scores (assessed before the tax changes) decreases, suggesting that there is advantegeous selection.

Mental status scores in the Rand harmonized longitudinal files from the Health and Retirement Survey are computed as the sum of vocabulary, naming, and counting scores from the HRS. Those scores are the sum of correct answers from questions ranging from "Who are the current president and vice-president of the United States?" to "How much is 100 minus 7? How much is that minus 7? [...]." The detailed construction of the mental status score variable is presented in Appendix C.4. Those measures can be seen as a proxy for ability, similarly to how Armed Forces Qualification Test (AFQT) scores in the National Longitudinal Survey of Youth is often used as a measure of ability (Farber and Gibbons, 1996; Altonji and Pierret, 2001; Lange, 2007; Craig, 2020). While the HRS scores are less detailed, they have the advantage of being assessed repeatedly for each respondent, at every survey year, and for that reason may be a more accurate measure of ability if ability is not constant but evolves dynamically over time.

Looking at the HRS total mental status scores as a proxy of ability, we regress those scores as measured two years before the baseline year, on changes in marginal retention[31] and a set of control variables (X_{it}, including year fixed effects, marital status and flexible functions of hourly wages), restricting the sample to those who are working in the baseline year and four years in the future. The coefficient η_r^m on regression equation 16 translates how different are the average mental status scores of those who are working after changes in marginal retention.

$$scores_{it} = \eta_r^m \Delta \log r_{it} + \gamma' X_{it} + u_{it} \tag{16}$$

The results for different sets of controls are presented in Table 1 in Appendix C.1. Fixing the set of controls, the results for different time horizons are presented in Figure 7 in Appendix C.1. For a horizon of four years after the baseline year, under the most stringent specification, we find that in response to a one percent increase in marginal retention between the baseline year and two years ahead, there is a change in the composition of the pool of workers such that average mental status scores (as measured before the tax change) decreases by 1.3 points (out of 15), with a standard deviation of 0.5.

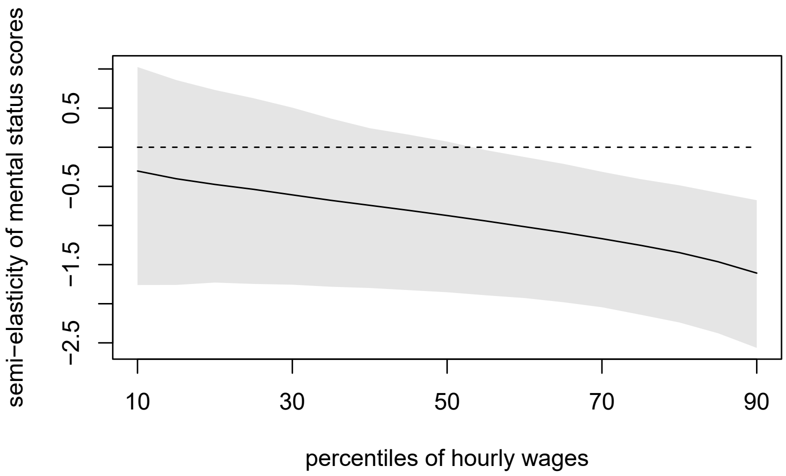
To estimate heterogeneous effects across different workers, we also estimate Equation 16 locally as a function of hourly salaries, which we treat as proxies for resumes,[32] using local polynomial methods.[33] The local results are presented in Figure 2. They show that semi-elasticities of mental status scores are higher in magnitude for high-earners.

[31]We focus on changes in marginal retention for three main reasons, as opposed to alternative measures such as accrual rates of Social Security wealth (Coile and Gruber, 2007), the average retention rates and their levels. First, retirement decisions do not necessarily take place at the beginning of the year, or require the worker to receive no taxable income in the future, so that within a year that decision can be though of as being similar to an intensive margin decision. Second, we want to capture compensated effects more than income effects, since we can unambigously predict the sign of the compensated effects. Finally, although in a dynamic context the changes in retirement incentives would be better captured by a complex combination of changes in accrual rates from Social Security wealth, expected taxes and salaries in the future, marginal tax rates are simple and easy to calculate with the help of the NBER tax simulator (Feenberg and Coutts, 1993).

[32]For evidence of further heterogeneity across education groups, occupations, and income levels, see Sztutman (2023), Chapter 1, Appendix 1.10.4.

[33]That is, for each hourly wage level y, we estimate the regression equation $scores_{it} = \eta_r^m(y)\Delta \log r_{it} + \beta(y)\Delta \log r_{it}(y_{it} - y) + \gamma(y)' X_{it} + u_{it}$, where observations are weighted by their distance from the hourly wages level y where the equation is being evaluated using the Epanechnikov kernel. An additional cross-term $\Delta \log r_{it}(y_{it} - y)$ is included to improve on the bias-variance tradeoffs, as explained in more detail in Fan and Gijbels (1996). Optimal bandwidths are selected with the leave-one-out cross validation procedure proposed by Racine (1993). Bootstrap confidence intervals are generated using the basic bootstrap method described in Chapter 5 of Davison and Hinkley (1997).

Figure 2: Semi-elasticities of mental status scores over different hourly wages percentiles



Notes. 95% bootstrap confidence intervals. Semi-elasticities are computed from linear regressions of memory scores two years in the past on changes in log marginal retention rates over two years (evaluated at the base year income data), restricting the sample to those who are working in the baseline year and two years ahead. Each regression includes year fixed effects, marital status dummies, and a 10-piece linear spline of log hourly wages.

5.2.2 Salary Changes

This section analyzes how salaries and participation rates respond to tax changes. By imposing further assumptions, namely that salaries are equal to the average productivity of workers, we will be able to quantify the importance of selection in labor markets from the point of view of optimal taxation.

The basic idea is that, under the assumption that salaries are equal to the average productivity of the workers, when we observe salaries before and after a tax increase, we can decompose average productivities at the initial period as the sum of the productivities of those who stay and of the marginal workers who dropped out of the labor force. That is, $\mathbb{E}[v_{before}] = \mathbb{E}[v_{after}](1 - s_{mg}) + v_{mg}s_{mg}$,where, $\mathbb{E}[v_{before}]$ and $\mathbb{E}[v_{after}]$ denotes the average productivity (or salaries) of the workers with a common resume, respectively before and after the tax increase; s_{mg} is the share of people who are induced to retire by the increase in marginal taxes, and v_{mg} is their productivity. From that observation, we can also establish a relationship between labor wedges (χ), elasticities of salaries (ϵ_r^w) and semi-elasticities of labor market participation (η_r^P):

$$\chi = \frac{v_{mg}}{\mathbb{E}[v_{after}]} = 1 - \frac{\frac{\mathbb{E}[v_{after}] - \mathbb{E}[v_{before}]}{\mathbb{E}[v_{after}]}}{s_{mg}} \implies \chi = 1 + \frac{\epsilon_r^w}{\eta_r^P} \tag{17}$$

To take the relationship given by Equation 17 to the data, we also have to address the usual identification concerns that come from estimating elasticities. Thus, we will estimate elastiticities of wages by regressing individual changes in log hourly salaries on simulated changes in the log of marginal retention rates, including different sets of controls X_{it} (year-fixed effects, initial hourly wages, and marital status),[34] as in Equation 18. The key identification assumption is that future productivity changes, and elasticities of participation, are independent of each other and of tax changes conditional on the set of controls.[35]

$$\Delta \log w_{it} = \epsilon^w \Delta \log r_{it} + \gamma' X_{it} + u_{it} \tag{18}$$

The results of this set of regressions are presented in Table 1 and Figure 5 in Appendix C.1. For the main specification, which includes year fixed effects, marital status dummies, and a 10-piece linear spline on hourly wages as controls, the estimated elasticity of wages is -0.16, with a standard deviation of 0.1, implying that a 1% increase in marginal retention between years 0 and 2 causes a 0.16% decrease in salaries between years 0 and 4. The effects are stronger at the 6-year horizon, with a point estimate of -0.27, and revert back at the 8-year horizon to -0.14, when estimates also get noisier. This is in line with the idea that marginal tax increases push the people who were almost indifferent between retiring or not into retirement, and those people are less productive than the average worker although they were receiving the same salaries. Salaries then would increase as employers realize that there was a change in the productivity composition of the pool of workers still on the labor force.[36]

To obtain estimates for the semi-elasticities of participation, analogously, we regress changes in participation[37] on simulated changes in the log of marginal retention rates,

[34]The controls aim to capture i) the possibility that changes in the tax schedule and in hourly wages may both respond to business cycle fluctuations (thus the inclusion of year-fixed effects), ii) the possibility that tax changes may have targeted different income groups and wages may evolve differently for those different groups (thus the inclusion of log hourly wages, and other non-linear functions of hourly wages), and iii) similarly, tax changes may have targeted differentially people of different marital status, for whom wages may evolve differentially as well (thus the inclusion of marital status indicator variables).

[35]This assumption becomes weaker when we estimate heterogeneous elasticities by different groups, as then the elasticities of participation need to be independent of future productivity changes only within groups.

[36]It is common wisdom that workers experience most of their salary changes when they change jobs. To more precisely capture the effects of changes in marginal tax rates on the wages of workers, we also consider the effects of restricting the sample to only those who switch jobs over the relevant time period, finding effects that are larger in magnitude, but less precisely estimated. See Sztutman (2023), Chapter 1, Appendix 1.10.4.

[37]That is, changes in the indicator variable that is equal to one whenever the individual is working.

including different sets of controls, as in Equation 19.[38]

$$\Delta p_{it} = \eta_r^p \Delta \log r_{it} + \gamma' X_{it} + u_{it} \tag{19}$$

The estimated semi-elasticities of participation when including the full set of controls and the 10-piece spline on hourly wages are of the order of 0.10 at the 2-year horizon, 0.01 at the 4-year horizon and 0.03 at the 6-year horizon, implying that at the 4-year horizon, a one percent increase in marginal retention causes a 1 percentage point decrease in the probability of a worker getting out of the labor force (see Figure 6 in Appendix C.1). The relatively larger effects at the shorter horizons when compared to the elasticity of wages is consistent with the idea that, after a tax increase, first, some of the workers drop out of the labor market, and then, as the employers learn that the pool of the remaining workers is more productive, wages gradually increase as time passes.

Our estimates are in line with a literature that has found that the implied tax rates on labor income from public pension rules have large disincentive effects on work (Gruber and Wise, 1998; Coile and Gruber, 2007). On the other hand, they are higher than the substitution elasticities inferred from variation in Social Security benefits from the 1977 Social Security Act that created the so called "Notch generation" Krueger and Pischke (1992); Gelber et al. (2016). While our confidence intervals are relatively wide, the difference can also be explained by the fact that the substitution incentives of their reform, and of Social Security benefits more generally, are relatively more opaque (Blinder et al., 1980) than the substitution incentives from income tax reforms.

Under the assumption that the elasticities are constant across different worker profiles – which will be relaxed below – we can take the ratio of the estimated coefficients $\frac{\epsilon_r^w}{\eta_r^P}$ (where η_r^P multiplies η_r^p by one hundred so that the numerator and denominator are in consistent units), to obtain a first set of estimates for the magnitude of the labor market informational externality $(1-\chi)$. These results are presented in Table 2 in Appendix C.1. Including the full set of controls and the 10-piece spline on hourly wages in both the participation and wages regressions, the estimated negative informational externality is around 0.16 when looking at effects over a 4-year horizon and around 0.015 when looking at participation effects over a 2-year horizon. In other words, workers would be paid around 1.5% to 16% more than their marginal productivity for the last unit of labor they supply.

As in the previous section, we estimate Equations 18 and 19 locally as a function of

[38]Again, the controls aim to capture the possibility that changes in the tax schedule and in labor market participation may both respond to business cycles fluctuations (thus the inclusion of year-fixed effects), the possibility that tax changes may have targeted different income groups and labor market participation may evolve differently for those different groups (thus the inclusion of log hourly wages, and other non-linear functions of hourly wages), and similarly, tax changes may have targeted differentially people of different marital status, for whom labor market participation may evolve differentially as well (thus the inclusion of marital status indicator variables).

hourly salaries, which we treat as proxies for resumes, using local polynomial methods.[39] Then, we use these estimates to compute labor wedges χ (Figure 3). Since elasticities of wages are higher in magnitude for high-earners, and high-earners also have lower participation semi-elasticities, labor wedges χ are decreasing in hourly wages, in line with the idea that informational imperfections are a larger issue for high-earning occupations and jobs. The point estimate for the elasticities of wages suggests that at the top of the distribution of hourly salaries the labor wedges could be very high, but also are imprecisely estimated. However, the values come from a combination of high elasticities of hourly wages and low participation elasticities, which approach zero, raising concerns about the validity of the bootstrap confidence intervals. The general message, however, is that we should expect the labor wedge χ to be decreasing across income levels, since they are coming from a combination of, as a function of income, increasing elasticities of wages and decreasing semi-elasticities of participation, which are robust features of the data.

Figure 3: Estimates for one minus the labor wedge for different hourly wages percentiles

Notes. 95% bootstrap confidence intervals. Estimates are computed from the ratio of elasticities of wages and semi-elasticities of participation. Elasticities of participation are computed from local linear regressions of changes in log hourly wages over four years on changes in log marginal retention rates over two years (evaluated at the base year income data). Semi-elasticities are computed from linear regressions of changes of hours wages over two years on changes in log marginal retention rates over two years (also evaluated at the base year income data). Each regression includes year fixed effects, marital status dummies, and a 10-piece linear spline of log hourly wages.

[39]That is, for each hourly wage level y, and dependent variable dep_{it},we estimate the regression equation $dep_{it} = \epsilon_r(y)\Delta \log r_{it} + \beta(y)\Delta \log r_{it}(y_{it} - y) + \gamma(y)' X_{it} + u_{it}$, where observations are weighted by their distance from the hourly wages level y where the equation is being evaluated using the Epanechnikov kernel. An additional cross-term $\Delta \log r_{it}(y_{it} - y)$ is included to improve on the bias-variance tradeoffs, as explained in more detail in Fan and Gijbels (1996). Optimal bandwidths are selected with the leave-one-out cross validation procedure proposed by Racine (1993). Bootstrap confidence intervals are generated using the basic bootstrap method described in Chapter 5 of Davison and Hinkley (1997). These results are presented in Figures 8 and 9 in Appendix C.1

5.3 Comparison to Existing Evidence

In this section, we compare our estimates to the available evidence on the time patterns of salaries and signaling on labor markets. We show that a back-of-the envelope calculation using the existing evidence from the literature would result in a similar magnitude for the Pigouvian component of taxes we found in the previous section.

There is documented evidence that workers experience large growth rates of salaries as a function of experience. In fact, Guvenen et al. (2021) have documented that the top 1% earners have a very steep growth rate of salaries, of around 2700% over a 30-year period, or 11.3% per year (and approximately 3% per year on average across workers). While part of this pattern may be thought as the result of human capital accumulation, and another part of it may be thought of as the result of pure luck, it is reasonable to expect that at least another part of it is due to selection and the career concerns logic we have uncovered. In fact, Guvenen et al. (2021) argue that no empirically plausible model of stochastic productivities could explain the large growth rates of salaries observed at the top. Moreover, there is evidence that signaling and learning are important to explain the dynamics of salaries and tenure in some occupations. For example, Cella et al. (2017) have shown that the relationship between the volatility of stock returns and the tenure of CEOs of large US firms is consistent with the idea that the market gradually learns about the CEO ability throughout the years of the CEO tenure.

For the purpose of this paper, the key question is determining the share of the growth rate of salaries that is due to signaling and how it can be translated into an estimate of the corrective component of taxes. However, there is no direct estimate in the literature that can readily be used to answer these questions. To get a sense of what would be reasonable magnitudes for the signaling component of the growth rate of salaries, we can look at the evidence on the return to schooling and the role of signaling in that context. Recent work has concluded that on average 30% of the returns to schooling can be attributed to signaling and 70% to human capital accumulation (Aryal et al., 2019). Combining the growth rate of salaries for top earners from Guvenen et al. (2021) with the signaling fraction of the return to schooling from Aryal et al. (2019), we can guess that the return to experience due to signaling, for top earners, may be of the order of 3.4% (30% of 11.3%) per year at 10 years of experience. Using Equation 3 (the free-entry condition for firms), we can show that the return to experience is related to the Pigouvian component of taxes $(1-\chi)$, and the shape of the lifetime income distribution by the formula $\gamma = \frac{\alpha_y(1-\chi)}{1+\alpha_y(1-\chi)}$. This implies that the Pigouvian component of taxes at the top could be as high as 25%, and on average around 6%.

6 Conclusion

While dynamic signaling is a key feature of labor markets, standard benchmark taxation models often ignore it, or adopt a static perpective on information transmission. In this paper, we developed a simple model that allows job histories and resumes to play this informational role, with firms using them to predict productivities and forward-looking individuals making labor supply decisions that anticipate the impact of these decisions today on future wages. The model builds a bridge between static models of the rat race and dynamic models of career concerns. In the model, the interest from firms in learning the productivity of workers arises even when firms are allowed to pay for performance and both workers and firms are risk neutral.

We derived generalized Mirrleesian and Pigouvian formulas that apply not only to this particular model of imperfect information in the labor market, but more generally to models with labor market frictions, as long as, given how firms set their wages, earnings map one-to-one to labor supply choices. The formulas decompose optimal taxes into two components: a redistributive component, and a corrective component. While a large non-linear income taxation literature has explored and estimated the statistics that appear in the redistributive component of taxes, there is limited work estimating the corrective component of taxes, especially in the context of dynamic imperfect information in labor markets. Using data from the Health and Retirement Study survey, this paper has shown that for an average worker, the corrective component of taxes is of the order of 5%, while for high earners it ranges from 10% to as high as 60%. This result has implications for the redistributive effects of the tax system. It implies that, once we take into account the impact of imperfect information in labor markets, we may conclude that the current tax system is less redistributive than we previously thought. Moreover, using the tax system to correct intertemporal distortions from the dynamic "rat race" could generate pure efficiency gains.

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Online Appendix

A Omitted Proofs

A.1 Algebra Behind Example in Section 3.1

Salaries are given by the expectation of productivity of those who provide at least h. In terms of types, salaries are given by $w(h(\theta)) = \mathbb{E}[v(\tilde{\theta})|\tilde{\theta} \geq \theta] = \frac{\alpha}{\alpha-\delta}\theta^{\delta}$. The first order conditions of the worker imply that:

$$w(h) = h^{1/\epsilon} b(\theta)^{-(1+1/\epsilon)} \tag{20}$$

Guessing and verifying that the wage is a power function, i.e., $w(h) = kh^{\gamma}$, for some k and γ, we conclude that:

$$w(h) = \frac{\alpha}{\alpha - \delta} k^{\frac{-\delta\epsilon}{(1+\epsilon)(1-\delta)}} h^{\frac{1-\epsilon\gamma}{(1+\epsilon)(1-\delta)}} \tag{21}$$

And therefore $w(h) = kh^{\gamma}$, where $k = \left(\frac{\alpha}{\alpha-\delta}\right)^{\frac{(1-\delta)(1+\epsilon)}{(1-\delta)(1+\epsilon)-\delta\epsilon}}$ and $\gamma = \frac{\delta}{1-\delta+\epsilon}$, as claimed in the main text.

A.2 Atkinson-Stiglitz with "Double Adverse Selection" (Proof of Proposition 1)

Proof. Any tax system generates a common budget set B that determines which pairs (C, H) are feasible. As a first step in the proof, we are going to show that we can replicate this budget set for the workers and save resources, without imposing taxes on timed consumption or labor flows. To do that, we generate a new tax system where, for each pair (C, H) the new pre-tax income is $e(H)$ and post-tax income is $e(C)$, where these are described by Equation 24 and Equation 23, respectively. Under this new tax system, the worker problem can be written in three parts:

$$\underset{C,H}{max}\ U(C, H, \theta)\ st.\ (C, H) \in B,\ \ B = \{C, H : e(H) \geq e(C)\} \tag{22}$$

$$e(C) = \underset{\tilde{c}}{min}\ \int_0^1 q(a) \cdot \tilde{c}(a) da\ st.\ C(\tilde{c}) \geq C \tag{23}$$

$$e(H) = \underset{\tilde{h}}{max}\ \int_0^1 q(a) \cdot w(h(a)) \cdot \tilde{h}(a) da\ st.\ H(\tilde{h}) \leq H \tag{24}$$

Notice that $e(H)$ is the maximum pre-tax income that can be generated by generating at most the disutility H. Moreover, it depends only on lifetime labor supply and not on the timing of these labor supply decisions, because $\int_0^1 q(a) \cdot w(h(a)) \cdot \tilde{h}(a) da = W(h)$.

Because production depends only on lifetime labor supply (and is increasing in lifetime labor supply), it is also the maximum production that generates at most the disutility H.

Further, because $e(C)$ is the smallest amount of resources that achieves the subutility level C, and $e(H)$ is the maximum production that can be generated by generating at most the disutility H, whenever choices change, there are more resources than required to obtain the same allocation. These extra resources can be used to increase all the consumption possibilities $(C + \Delta(H), H)$ by some small amount $\Delta(H)$, chosen in such a way that resulting allocation is incentive compatible and the resource contraint is still satisfied.

□

A.3 Invertibility Condition (Proof of Lemma 2)

Lemma. *Without loss, we can solve directly for $\tilde{R}(h) = R(y(h))$, and then find $y(h)$, and $R(y)$. That is, the planner can solve the simpler problem:*

$$\max_{\tilde{R}(h)} \mathbb{E}[W(V(\tilde{R}, \theta), \theta)] \ s.t. \ \mathbb{E}[v(\theta)h(\tilde{R}, \theta) - \tilde{R}(h(\tilde{R}, \theta))] \geq 0 \tag{25}$$

Proof. Because $y'(h)$ is a well-defined function of the allocation and is always positive, $y(h)$ always exists and $y(h)$ is strictly increasing. Thus, there exists an inverse function $h^{-1}(y)$. Therefore, we can define $R(y)$ so that $R(y(h)) = \tilde{R}(h)$. Thus, we found the income tax schedule and equilibrium salaries that prevail in the economy where the planner solved directly for the retention function $\tilde{R}(h)$. □

A.4 Optimal Taxes with Single Dimensional Heterogeneity (Proof of Proposition 2)

Proof. First, we derive a necessary condition for optimality of the planning problem written in terms of post-tax wages, where the planning problem is given by:

$$\max_{\tilde{R}} \mathbb{E}[W(V(\tilde{R}; \theta), \theta)] \ s.t. \ \mathbb{E}[v(\theta)h(\tilde{R}, \theta) - \tilde{R}(h(\tilde{R}, \theta))] \geq 0 \tag{26}$$

Define $\mathcal{L}(\tilde{R}) = \mathbb{E}[W(V(\tilde{R}; \theta), \theta)] - \mu \mathbb{E}[v(\theta)h(\tilde{R}, \theta) - \tilde{R}(h(\tilde{R}, \theta))]$, where μ is the Lagrange multiplier on the resource constraint. Let's consider a small one-bracket variation on marginal retention rates as a function of effort at a given level of effort, i.e., let's consider the consequences of changing the tax schedule to $\tilde{R}^* = \tilde{R} + \epsilon\Delta$, where

$$\Delta(x, \ell) = \begin{cases} 0 & \text{if } x \leq h \\ x - h & \text{if } h \leq x \leq h + \ell \\ \ell & \text{if } x \geq h + \ell \end{cases} \tag{27}$$

.

For any functional $\mathcal{F}$, let $\frac{d\mathcal{F}}{d\tilde{r}_h} \equiv \underset{\ell \to 0^+}{lim} \frac{1}{\ell} \frac{d}{d\epsilon}\Big|_{\epsilon=0} \mathcal{F}(\tilde{R}^*)$. Moreover, denote $\frac{d\mathcal{F}}{dI} \equiv \frac{d}{d\epsilon}\Big|_{\epsilon=0} \mathcal{F}(\tilde{R}+\epsilon\Delta_I)$, where $\Delta_I(x) = 1, \forall x$. If the tax schedule is optimal, then, changing the tax schedule towards the direction Δ should not increase the value of the Lagrangian, i.e.:

$$\frac{d\mathcal{L}}{d\tilde{r}_h} \equiv \underset{\ell \to 0^+}{lim} \frac{1}{\ell} \frac{d}{d\epsilon}\Big|_{\epsilon=0} \mathbb{E}[W(V(\tilde{R}^*, \theta), \theta)] + \mu \mathbb{E}[v(\theta) h(\tilde{R}^*, \theta) - \tilde{R}^*(h(\tilde{R}^*, \theta))] = 0 \tag{28}$$

Denote $\lambda(\theta) = \frac{dW(V,\theta)}{dV}$,and moreover, under the assumption that lifetime labor responds at the intensive margin, Equation 28 implies:

$$\mathbb{E}\Big[\lambda(\theta) \frac{dV(\tilde{R}, \theta)}{d\tilde{r}_h}\Big] = -\mu \mathbb{E}\Big[v(\theta) \frac{dh(\theta)}{d\tilde{r}_h} - \tilde{r}(h) \frac{dh(\theta)}{d\tilde{r}_h} - 1(h(\theta) \geq h)\Big] \tag{29}$$

Let's define $\frac{dh^c(\theta)}{d\tilde{r}_h} \equiv \frac{dh(\theta)}{d\tilde{r}_h} 1(h(\theta) = h)$, so that we can separate behavioral effects into income and compensating effects:

$$\begin{aligned}\mathbb{E}\Big[\frac{\lambda(\theta)}{\mu} \frac{dV(\tilde{R}, \theta)}{dI} 1(h(\theta) \geq h)\Big] = -\mathbb{E}\Big[&(v(\theta) - \tilde{r}(h(\theta))) \frac{dh^c(\theta)}{d\tilde{r}_h} \\ &- (v(\theta) - \tilde{r}(h(\theta))) \frac{dh(\theta)}{dI} 1(h(\theta) \geq h) - 1(h(\theta) \geq h)\Big]\end{aligned} \tag{30}$$

Denote $f(h)$ the density of lifetime labor h, and denote $v(h)$ the productivity of the worker supplying lifetime labor h, and ommiting the dependence on types:

$$\Big(v(h) - \tilde{r}(h)\Big) f(h) \frac{dh^c}{d\tilde{r}_h} = \int_h^\infty f(\tilde{h}) \Big(1 - \lambda(\tilde{h})\Big) d\tilde{h} + \int_h^\infty \Big(v(\tilde{h}) - \tilde{r}(\tilde{h})\Big) f(\tilde{h}) \frac{d\tilde{h}}{dI} d\tilde{h} \tag{31}$$

Now, writing it in terms of compensated and income elasticities, $\epsilon^c_{\tilde{r}}(h) \equiv \frac{dh^c}{d\tilde{r}_h} \frac{\tilde{r}(h)}{h}$ and $\eta_I(h) \equiv \frac{dh}{dI} \frac{I}{h}$:

$$\Big(\frac{v(h) - \tilde{r}(h)}{\tilde{r}(h)}\Big) f(h) h \epsilon^c_{\tilde{r}}(h) = \int_h^\infty f(\tilde{h}) \Big(1 - \lambda(\tilde{h})\Big) d\tilde{h} + \int_h^\infty \Big(\frac{v(\tilde{h}) - \tilde{r}(\tilde{h})}{\tilde{R}(\tilde{h})}\Big) f(\tilde{h}) \eta_I(\tilde{h}) d\tilde{h} \tag{32}$$

Now we are going to rewrite this formula in terms of lifetime income. Let $\frac{d\mathcal{F}}{dr}$ be the analogous derivative with respect to a change in retention as function of lifetime income (instead of lifetime labor supply). Notice that

$$\tilde{r}(h) = r(y(h)) y'(h) \tag{33}$$

And therefore,

$$\frac{dy(h)}{dr} = y'(h) \frac{dh}{d\tilde{r}_h} \frac{d\tilde{r}_h}{dr} = y'(h)^2 \frac{dh}{d\tilde{r}} \tag{34}$$

Additionally, we can relate y and h, their densities, and their income elasticities by the following equations:

$$f(h) = g(y(h)) y'(h) \tag{35}$$

$$\frac{dy(h)}{dI} = y'(h) \frac{dh}{dI} \tag{36}$$

$$dh = \frac{1}{y'(h)} dy \tag{37}$$

Substituting Equations 33 - 37 into Equation 32:

$$\left(\frac{v(h)/y'(h) - r(y(h))}{r(y(h))}\right) g(y(h)) y(h) \epsilon_r^c(y(h)) = \int_h^\infty f(\tilde{h}) \left(1 - \lambda(\tilde{h})\right) d\tilde{h} + \int_h^\infty \left(\frac{v(\tilde{h})/y'(\tilde{h}) - r(y(\tilde{h}))}{R(y(\tilde{h}))}\right) f(\tilde{h}) \eta_I(y(\tilde{h})) d\tilde{h} \tag{38}$$

Which can be read as a function of lifetime income, where $\chi(y) \equiv v(y)/y'(h(y))$:

$$\left(\frac{\chi(y) - r(y)}{r(y)}\right) g(y) y \epsilon_r^c(y) = \int_y^\infty g(\tilde{y}) \left(1 - \lambda(\tilde{y})\right) d\tilde{y} + \int_y^\infty \left(\frac{\chi(\tilde{y}) - r(\tilde{y})}{R(\tilde{y})}\right) g(\tilde{y}) \eta_I(\tilde{y}) d\tilde{y} \tag{39}$$

□

A.5 Conditioning on the "Strength of the Resume" (Proof of Proposition 3)

In this section, we assume that signals take the form of $h_\phi = I(\tilde{h}, a, \theta) = \int_0^a \phi(\tilde{a}, a) \tilde{h}(\tilde{a}) d\tilde{a}$, with $\phi(\tilde{a}, a) > 0$, continuous in a and $\tilde{a}$.

Using the logic from Proposition 2 and the assumption of homogeneous preferences over the timing of labor supply and consumption as in Proposition 1, we will show that optimal taxation formulas will still take a simple structure.

We first set aside the issue of implementation, and characterize the set of optimal incentive compatible allocations. Those allocations need to satisfy two properties, as stated in Lemma 3.

Lemma 3. *Any optimal and incentive compatible allocation satisfies two properties:*

Efficient timing: For any H and C,

$$\tilde{h}(\cdot; H)_0^1 = argmax \int_0^1 q(a) \tilde{h}(a) da \ st.\ H(\tilde{h}) = H \tag{40}$$

$$\tilde{c}(\cdot; C)_0^1 = argmin \int_0^1 q(a) \tilde{c}(a) da \ st.\ C(\tilde{c}) = C \tag{41}$$

Lifetime optimality:

$$\left(\frac{v(\theta) - \tilde{r}(e_H)}{\tilde{r}(e_H)}\right) f(e_H) e_H \epsilon_{\tilde{r}}^c(e_H) = \int_{e_H}^\infty f(e_H) \left(1 - \lambda(\tilde{e_H})\right) d\tilde{e_H} + \int_{e_H}^\infty \left(\frac{v(\theta) - \tilde{r}(\tilde{e_H})}{\tilde{R}(\tilde{e_H})}\right) f(\tilde{e_H}) \eta_I(\tilde{e_H}) d\tilde{e_H} \tag{42}$$

where $\tilde{R}(e_H) = e_C$, *and*

$$e_H = e(H) = \underset{\tilde{h}}{max} \int_0^1 q(a)\tilde{h}(a)da \ st. \ H(\tilde{h}) = H \tag{43}$$

$$e_C = e(C) = \underset{\tilde{c}}{min} \int_0^1 q(a)\tilde{c}(a)da \ st. \ C(\tilde{c}) = C \tag{44}$$

Proof. Efficient timing follows from an analogous argument to the production efficiency theorem of Diamond and Mirrlees (1971) (see also Lemma 7, below).

That is, consider the planning problem

$$\begin{aligned} &\underset{\tilde{c}_\theta, \tilde{h}_\theta}{max} \mathbb{E}[W(\mathrm{U}(C(\tilde{c}_\theta), H(\tilde{h}_\theta), \theta), \theta)] \\ &s.t. \mathbb{E}[v(\theta) \int_0^1 q(a)(\tilde{h}_\theta(a) - \tilde{c}_\theta(a))da] \geq 0 \\ &\quad U(C(\tilde{c}_\theta), H(\tilde{h}_\theta), \theta) \geq U(C(\tilde{c}_{\theta'}), H(\tilde{h}_{\theta'}), \theta) \ \forall \theta, \theta' \end{aligned} \tag{45}$$

We claim that any solution of this problem is a solution to the problem below, where we divide it into two parts, one where we solve for $C_\theta = C(\tilde{c}_\theta)$, and $H_\theta = H(\tilde{h}_\theta))$, and the other where we think of the optimal way to achieve each (C_θ, H_θ) pair.

$$\begin{aligned} &\underset{C_\theta, H_\theta}{max} \ \mathbb{E}[W(\mathrm{U}(C_\theta, H_\theta, \theta), \theta)] \ st \ \mathbb{E}[v(\theta)e(H_\theta) - e(C_\theta)] = 0 \\ &\qquad\qquad U(C_\theta, H_{\theta'}, \theta) \geq U(C_{\theta'}, H_{\theta'}, \theta) \ \forall \theta, \theta' \end{aligned} \tag{46}$$

Where $e(H_\theta)$ and $e(C_\theta)$ satisfy Equations 43 and 44, with $H = H_\theta$, and $C = C_\theta$.

Why? Suppose it is not. Then, fix original C_θ, H_θ, but set $\tilde{c}_\theta = \tilde{c}(\cdot; C_\theta)_0^1$ and $\tilde{h}_\theta = \tilde{h}(\cdot; H_\theta)_0^1$ as the solution to the sub problems 40 and 41. This does not affect the incentive compatibility constraints, but saves resources. That implies the resource constraint is slack and we can find a better solution. In particular, let $\tilde{R}$ to be the implicit retention map generated by the original candidate solution, i.e, $C_\theta = \tilde{R}(H_\theta)$. We can increase $\tilde{R}$ uniformly, and for a sufficiently small increase, given the continuity of worker choices, the resource constraint will still be satisfied. This generates a Pareto improvement, and improves the objective function of the planner, contradicting the optimality of the original C_θ, H_θ.

Now to prove the second part of the claim ("lifetime optimality"), we proceed to make a standard variational argument over the retention schedule $\tilde{R}$, for the problem 47 defined below, where, $V(\tilde{R}, \theta) = \underset{e_H}{max} \tilde{U}(\tilde{R}(e_H), e_H, \theta)$, $e_H(\tilde{R}, \theta) = \underset{e_H}{argmax} \tilde{U}(\tilde{R}(e_H), e_H, \theta)$, $\tilde{U}(e_C, e_H, \theta) = U(C, H, \theta)$, and $e_C = \tilde{R}(e_H)$.

$$\underset{\tilde{R}}{max} \ \mathbb{E}[W(\mathrm{V}(\tilde{R}, \theta), \theta)] \ s.t. \ \mathbb{E}[v(\theta)e_H(\tilde{R}, \theta) - \tilde{R}(e_H(\tilde{R}, \theta))] = 0 \tag{47}$$

Assuming that no bunching takes place at the optimal assignment, and following the steps from the proof of Proposition 2, we conclude that the following condition is necessary:

$$\left(\frac{v(\theta)-\tilde{r}(e_H)}{\tilde{r}(e_H)}\right)f(e_H)e_H\epsilon^c_{\tilde{r}}(e_H) = \int_{e_H}^{\infty} f(e_H)\Big(1-\lambda(\tilde{e_H})\Big)d\tilde{e_H} + \int_{e_H}^{\infty}\left(\frac{v(\theta)-\tilde{r}(\tilde{e_H})}{\tilde{r}(\tilde{e_H})}\right)f(\tilde{e_H})\eta_I(\tilde{e_H})d\tilde{e_H}. \tag{48}$$

□

Lemma 4. *Assuming* $\frac{d^2H(\tilde{h}(\cdot)_0^1)}{d\tilde{h}(a)d\tilde{h}(a')} < 0$, $MRS(C,H,\theta) \equiv -\frac{U_H(C,H,\theta)}{U_C(C,H,\theta)}$ *decreasing in* θ, $\mathtt{I}(\tilde{h},a,\theta) = \int_0^a \phi(a,\tilde{a})\tilde{h}(\tilde{a})$, *salaries are increasing in* $\mathtt{I}$.

Proof. Because $MRS(C,H,\theta)$ is decreasing in θ, for any $\tilde{R}(H)$ strictly increasing – which is a property of the optimal allocation described in Lemma 3 – higher types θ choose higher levels of H. Since $\frac{d^2H(\tilde{h}(\cdot)_0^1)}{d\tilde{h}(a)d\tilde{h}(a')} < 0$, higher levels of H imply higher levels of each $\tilde{h}(a)$. Therefore, conditional on age, higher types will choose strictly higher indexes, and thus for any $\mathtt{I} > \mathtt{I}'$ and $a > 0$, $\mathbb{E}[v(\theta)|\mathtt{I},a] > \mathbb{E}[v(\theta)|\mathtt{I}',a]$. Therefore, $\mathbb{E}[v(\theta)|\mathtt{I}] > \mathbb{E}[v(\theta)|\mathtt{I}']$, and salaries are increasing in $\mathtt{I}$. □

Lemma 5. *No two different continuous functions* $\tilde{h}$ *and* $\tilde{h}'$ *map into the same* $\tilde{y}$.

Proof. Remember that $\tilde{y}(a;\tilde{h}^a) = \tilde{h}(a)\cdot w(\tilde{h}^a) = \tilde{h}(a)\cdot\mathbb{E}[v(\theta)|\mathtt{I}(\tilde{h}^a)]$. Consider the first positive measure interval where $\tilde{h}$ and $\tilde{h}'$ differ, and without loss consider an interval $(\underline{a},\bar{a})$ where $\tilde{h}(a) > \tilde{h}'(a)$. If salaries were the same for both functions, in that interval $\tilde{y}(a;\tilde{h}^a) > \tilde{y}'(a,\tilde{h}'^a)$, and the proof would be complete. However, by Lemma 4 salaries are increasing as a function of $\tilde{h}(a)$, for any a; thus in this case we also have that $\tilde{y}(a;\tilde{h}^a) > \tilde{y}(a,\tilde{h}'^a)$. Therefore the different functions $\tilde{h}$ and $\tilde{h}'$ map into the different earnings $\tilde{y}$ and $\tilde{y}'$. □

Lemma 5 allows us to decentralize any allocation with the properties from Lemma 3.

Proposition 5. *If* $R(\tilde{y})$ *is optimal then there are* R_m, R_p *with* $R(\tilde{y}) = R_m(R_p(\tilde{y}))$, *such that* R_m *and* R_p *satisfy the following conditions:*

$$v(\tilde{h})q(a) = \frac{dR_p(\tilde{y})}{d\tilde{y}(a)}w(\tilde{h}^a) + \int_a^1 \frac{dR_p(\tilde{y})}{d\tilde{y}(\tilde{a})}\frac{dw(\tilde{h}^{\tilde{a}})}{d\tilde{h}(a)}\tilde{h}(\tilde{a})d\tilde{a}, \tag{49}$$

$$\left(\frac{1-r_m(R_p)}{r_m(R_p)}\right)g(R_p)R_p\epsilon^c_{\tilde{r}}(R_p) = \int_{R_p}^{\infty} g(\tilde{R}_p)\Big(1-\lambda(\tilde{R}_p)\Big)d\tilde{R}_p + \int_{R_p}^{\infty}\left(\frac{1-r_m(\tilde{R}_p)}{r_m(\tilde{R}_p)}\right)g(\tilde{R}_p)\eta_I(\tilde{R}_p)d\tilde{R}_p, \tag{50}$$

Proposition 5 states that the tax system should be such that i) history-dependent taxes (R_p) should be used to correct for labor wedges, and ii) after correcting for these distortions, lifetime income redistributive taxes should be imposed on top of these taxes, according to standard redistributive formulas.

Where $v(\tilde{h})$ is the marginal productivity of the type that supplies $\tilde{h}$, and $w(\tilde{h}^a)$ is the salary for someone with a history of labor supply $\tilde{h}^a$. For ease of notation the dependence on $\tilde{y}$ is omitted. That is, the formula should be read as a function of earnings flows $\tilde{y}$, through the inverse operator $\tilde{h}(\tilde{y})$.

Proof. We have established that we can find the optimal post-tax wages $\tilde{R}$, by solving the following problem:

$$\underset{\tilde{R}}{max}\ \mathbb{E}[W(V(\tilde{R},\theta),\theta)] \text{ s.t. } \mathbb{E}\left[\int_0^1 v(\theta)q(a)\tilde{h}(a;\tilde{R},\theta)da - \tilde{R}(\tilde{h}(\cdot;\tilde{R},\theta))\right] = 0 \tag{51}$$

That is, we solve for the optimal post-tax wages as if there were a first layer of taxes guaranteeing that the lifetime gains from increasing the labor supply are equal to the contribution to output, i.e.:

$$\frac{\partial \tilde{R}_p(\tilde{h})}{\partial \tilde{h}(a)} = v(\tilde{h})q(a) \tag{52}$$

In terms of earnings, we have that:

$$R_p(\tilde{y}(\tilde{h})) = \tilde{R}_p(\tilde{h}) \tag{53}$$

Therefore,

$$\frac{\partial \tilde{R}_p(\tilde{h})}{\partial \tilde{h}(a)} = \frac{dR_p(\tilde{y})}{d\tilde{y}(a)}w(\tilde{h}^a) + \int_a^1 \frac{dR_p(\tilde{y})}{d\tilde{y}(\tilde{a})}\frac{dw(\tilde{h}^{\tilde{a}})}{d\tilde{h}(a)}\tilde{h}(\tilde{a})d\tilde{a} \tag{54}$$

and

$$v(\tilde{h})q(a) = \frac{dR_p(\tilde{y})}{d\tilde{y}(a)}w(\tilde{h}^a) + \int_a^1 \frac{dR_p(\tilde{y})}{d\tilde{y}(\tilde{a})}\frac{dw(\tilde{h}^{\tilde{a}})}{d\tilde{h}(a)}\tilde{h}(\tilde{a})d\tilde{a}, \tag{55}$$

To arrive at the special case with no intertemporal distortions, notice that having no (pre-tax) intertemporal distortions implies that $\frac{dR_p(\tilde{y})}{d\tilde{y}(\tilde{a})}q(\tilde{a}) = \frac{dR_p(\tilde{y})}{d\tilde{y}(a)}q(a)$ for every $\tilde{a}$ and a, and therefore the equation above simplifies to

$$\frac{dR_p(\tilde{y})}{d\tilde{y}(a)}\frac{1}{q(a)} = \frac{v(\tilde{h}^a)q(a)}{q(a)w(\tilde{h}^a) + \int_a^1 \frac{dw(\tilde{h}^{\tilde{a}})}{d\tilde{h}(a)}\tilde{h}(\tilde{a})q(\tilde{a})d\tilde{a}} \tag{56}$$

□

Proposition. *If resumes are summarized by their "pace," under the parametric assumptions of Section 3.2, then* $\frac{dR_p(\tilde{y})}{d\tilde{y}(a)} = a^\gamma e^{-\rho a}$*, where* $\gamma = \frac{\delta}{1+\epsilon-\delta}$*.*

Proof. We have established that in the "pace-of-the-resume" case, for each worker, labor supply is constant over time, pre-tax salaries depend only on the pace $\bar{h}$, and in equilibrium are equal to their productivities. Denote the pace at age a as $\bar{h}(a) = \int_0^a \frac{\tilde{h}(\tilde{a})}{a}d\tilde{a}$. Equation 14 can be written as:

$$w(\bar{h})e^{-\rho a} = \frac{dR_p(\tilde{y})}{d\tilde{y}(a)} w(\bar{h}(a)) + \int_a^1 \frac{dR_p(\tilde{y})}{d\tilde{y}(\tilde{a})} \frac{dw(\bar{h}(\tilde{a}))}{d\tilde{h}(a)} \bar{h} d\tilde{a} \tag{57}$$

Using the fact that $\frac{dw(\bar{h})}{d\tilde{h}(a)} = w'(\bar{h}(\tilde{a}))\frac{d\bar{h}(\tilde{a})}{d\tilde{h}(a)} = \frac{w'(\bar{h})}{\tilde{a}}$, and defining the function $r_y(a)$ so that $\frac{dR_p(\tilde{y})}{d\tilde{y}(a)} = r_y(a)e^{-\rho a}$, we can rewrite it as:

$$e^{-\rho a} = r_y(a)e^{-\rho a} + \int_a^1 r_y(\tilde{a})\frac{e^{-\rho\tilde{a}}}{\tilde{a}}\frac{w'(\bar{h})\bar{h}}{w(\bar{h})} d\tilde{a} \tag{58}$$

Notice that under the parametric assumptions from Section 3.2, $\gamma = \frac{w'(\bar{h})\bar{h}}{w(\bar{h})} = \frac{\delta}{1+\epsilon-\delta}$. Taking a derivative with respect to a, we conclude that:

$$r_y'(a) = \gamma\ \frac{r_y(a)}{a} \tag{59}$$

And therefore $r_y(a) = a^\gamma$, and $\frac{dR_p(\tilde{y})}{d\tilde{y}(a)} = a^\gamma e^{-\rho a}$. □

A.6 Welfare (Proof of Proposition 4)

Proposition 6. *If the original tax schedule is optimal and the planner has redistributive preferences, workers' preferences are convex over leisure and consumption, and leisure is a normal good, then decreasing information asymmetries in labor markets decreases welfare.*

Definition. We say the planner has redistributive preferences if $W(V,\theta) = W(V)$ and $W'(V)U_c$ is positive and decreasing.

Proof. We can set the planner's problem as maximizing a welfare function of workers utility, subject to incentive compatibility constraints and a feasibility constraint.

$$\begin{aligned} \max_{l(\theta),c(\theta)} \ & \int W(u(c(\theta),l(\theta)))f(\theta)d\theta \\ s.t.\ & u\Big(c(\theta),l(\theta)\Big) \geq u\Big(c(\theta'),l(\theta')\frac{b(\theta')}{b(\theta)}\Big)\ \forall\theta,\theta'\ \text{[IC's]} \\ & \int (c(\theta) - v(\theta)b(\theta)l(\theta)f(\theta)d\theta \leq 0\ \text{[Feasibility]} \end{aligned}$$

The incentive compatibility constraints translate the idea that under the results from Lemma 2 (or the more general Lemma 4) the planner can solve for the allocation in terms of consumption and deliverables $h(\theta)$ subject to workers not being willing to misreport their types and trade their allocation $(c(\theta),h(\theta))$ to another allocation $(c(\theta'),h(\theta'))$. Moreover, the amount of deliverables $h(\theta)$ is equal to the product of effort $l(\theta)$ and observable component of productivities $b(\theta)$, i.e. $l(\theta)\cdot b(\theta) = h(\theta)$. Therefore the incentive compatibility constraints can be stated as requiring for each $\theta$and θ' that $u\Big(c(\theta),l(\theta)\Big) \geq u\Big(c(\theta'),l(\theta')\frac{b(\theta')}{b(\theta)}\Big)$.

In our context, the results in Seade (1982); Werning (2000) imply that out of the set of IC's, only the local downward are binding, provided that $W'(U)U_c$ is positive and decreasing, and that leisure is a normal good (they also further imply $v-r$ is positive. For more details, see Appendix B.5 and the argument in Werning (2000)). Those downward incentive compatibility constraints become tighter whenever informational asymmetries decrease, while the feasibility constraint is unchanged. Thus, welfare decreases.[40] We now walk through each of the steps of the argument.

First, we reproduce Werning's argument succintly below, that local IC's bind downwards, i.e. the lagrange multiplier on the local IC's from the planner's problem below are positive. First, we replace global incentive compatibility contraints with the local incentive compatibility constraints, and set up the planner's problem above as:

$$\max_{u,h} \int W(u(\theta))f(\theta)d\theta \tag{60}$$

$$s.t.\ u'(\theta) = U_\theta(e(u(\theta), h(\theta), \theta), h(\theta), \theta) \tag{61}$$

$$\int (e(u(\theta), h(\theta), \theta) - v(\theta)h(\theta))f(\theta)d\theta \leq 0 \tag{62}$$

We write the Lagrangian and, for conciseness, we drop the explicit dependence on θ

$$\max_{u,h} \int W(u)\ f + \mu(u' - U_\theta) - \kappa(e - vh)f\ d\theta \tag{63}$$

Integrate by parts

$$\max_{u,h} \int W(u)\ f - \mu' u - \mu U_\theta - \kappa(e - vh)f\ d\theta + u\mu|_{\underline{\theta}}^{\bar{\theta}} \tag{64}$$

Take FOC's for $u(\theta)$, and $h(\theta)$, denote $\lambda(\theta) = W'(u(\theta))$, and also note that $\mu(\bar{\theta}) = \mu(\underline{\theta}) = 0$.

$$\lambda f - \mu' - \mu U_{\theta,c}e_u - \kappa e_u f = 0 \tag{65}$$

$$\mu(U_{\theta,c}e_h + U_{\theta,h}) - ke_h f + kvf = 0 \tag{66}$$

Equation 66 can be rewritten as:

$$\mu U_c/\kappa = \frac{f\ \frac{v-r}{r}}{\frac{\partial \log MRS}{\partial \theta}} \tag{67}$$

Now we show that $\mu \geq 0$. Suppose towards a contradiction that there is a θ such that

[40]The result does not depend on the particulars of the information structure. Besides holding in the MWS case (as in Stantcheva (2014)), and in other models of competition with imperfect information such as the competitive screening model in Azevedo and Gottlieb (2017), and in the case of our model where resumes are defined as the length of the resume, it also holds under the more general information structure of Section 4.2, where resumes take the general form $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a \phi(\tilde{a}, a)\tilde{h}(\tilde{a})d\tilde{a}$.

$\mu(\theta) < 0$. Then, there is $\theta_1 < \theta_2$ such that $\mu(\theta_1) = \mu(\theta_2) = 0$, and $\mu'(\theta_1) < 0$, $\mu'(\theta_2) > 0$. Thus from Equation 67, we conclude that for $v(\theta_1) = r(\theta_1)$ and $v(\theta_2) = r(\theta_2)$, that is, they are unconstrained. Thus, it is also true that $e_u = 1/U_c$ for θ_1, and θ_2. Equation 65 tell us that when $\mu = 0$:

$$e_u = \frac{1}{\kappa}\Big(\lambda - \frac{\mu'}{f}\Big) \tag{68}$$

Therefore, $e_u(\theta_1) \geq e_u(\theta_2)$,and $U_c(\theta_2) \geq U_c(\theta_1)$. Under the assumption that preferences are convex and that leisure is a normal good, that would imply that $u(\theta_2) \leq u(\theta_1)$. That is, however, impossible: Since θ_2 is the more productive worker, that would violate their incentive compatibility constraints. Therefore, $\mu \geq 0$.

Now, we claim that a decrease in information asymmetries as we have defined them, makes the local incentive compatibility constraints tighter, and therefore reduces welfare. Notice that we can rewrite the resource constraint (Inequality 62) as:$\int(e(u(\theta), \ell(\theta)) - \theta\ell(\theta))f(\theta)d\theta \leq 0$ and the local incentive compatibility constraints (Equation 61) as $u'(\theta) = -\frac{\partial U(e(u(\theta),\ell(\theta)),\ell(\theta))}{\partial \ell}\ell(\theta)\frac{b'(\theta)}{b(\theta)} = -\frac{\partial U(e(u(\theta),\ell(\theta)),\ell(\theta))}{\partial \ell}\ell(\theta\Big(\frac{b_0'(\theta)}{b_0(\theta)} + \frac{\Delta'(\theta)}{\Delta(\theta)}\Big)$. The first is not affected by a change in information asymmetries as we have defined them, and the second constraint becomes tighter whenever information asymmetries decrease. Therefore, welfare decreases whenever information asymmetries decrease.

□

B Additional Theoretical Results

B.1 General Positive Properties: Strength of the Resume

This section shows properties of salaries and the return to experience that hold under a broader set of assumptions than those from the examples of Section 3. We focus on the case where the resume is summarized by a signal of the form $\mathtt{I}(\tilde{h}, a, \theta) = \int_0^a \tilde{h}(\tilde{a})\phi(\tilde{a}, a)d\tilde{a}$, with $\phi(\tilde{a}, a) > 0$, bounded and continuous. In contrast to the previous cases, where the signal was referred to the "length-of-the-resume" and the "pace-of-the-resume," we say that in this more general case the firms receive a public signal which can be called the "strength-of-the-resume" of the candidate.

The first general property is that, for any preferences and distribution of productivities, every worker starts their career with nothing to show in their resumes, and therefore the strength of their resumes is zero. Therefore, everyone has the same initial salary, which is equal to the average productivity of all workers. In particular, workers with above-average productivity start their career earning less than their marginal product, as the average productivity is everyone's initial salary. All workers have to climb the same career ladder and before they can show anything in their resume they are all indistinguishable for employers. This is illustrated in the left panel of Figure 4, where the trajectories of the

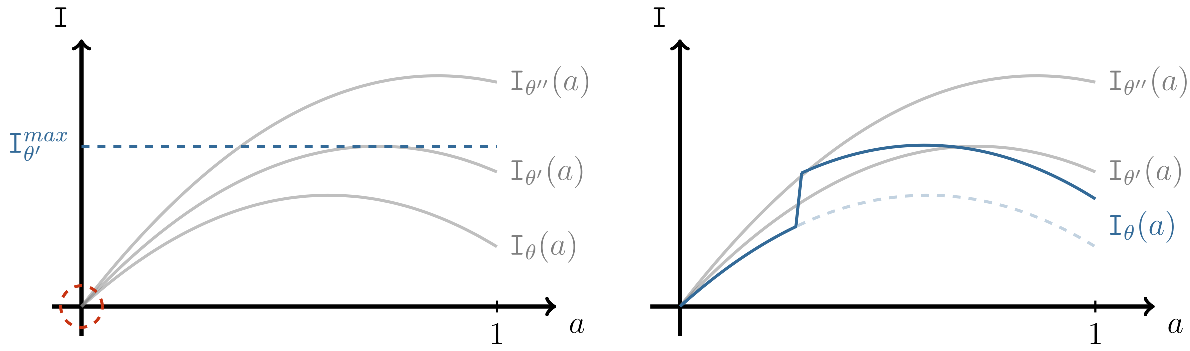


Figure 4: Career of Workers. Left Panel: Initial Salaries, and Salaries at the Peak. Right Panel: Signaling Return to Experience

"strength-of-the-resume" of different hypothetical workers are plotted as a function of their age, and the pool of workers who just started their careers is highlighted in red.

Second, workers at the peak of their careers (when they reach the highest value of the index $\int_0^a \tilde{h}(\tilde{a})\phi(\tilde{a}, a)d\tilde{a})$ will earn more than their marginal productivity because they will be pooled together only with the types that are willing to provide higher amounts of deliverables across all periods. This is illustrated in the left panel of Figure 4, where a dashed horizontal line shows the pool of workers who at some point in their careers have a resume of a certain "strength." Whenever this line is tangent to the highest "strength-of-the-resume" a worker achieves, this worker will be pooled together only with people who reach even higher "strength" levels and who are more productive than this worker (under the assumption that those who are more productive are more willing to supply deliverables). The left panel of Figure 4 raises the possibility that for any individual their resume may deteriorate over time. This could appear as the result of a worker failing to meet the expected pace of completion of deliverables. For example, in the "pace-of-the-resume" case from the Section 3.2, if a worker decided to reduce the pace of completion of deliverables, then the resume would become weaker; that would be interpreted as if it was bad news about the productivity of the worker.[41]

Third, at any point, assuming more productive types are more willing to provide the deliverables at all periods, exerting more effort increases future salaries. This is the case because whenever workers decide to exert more effort and improve their resumes, they will show to employers resumes that are more similar to the resumes of the more productive types. This is illustrated in the right panel of Figure 4, where workers of a certain type move from the dashed blue line to the solid blue line when they decide to exert more effort at a certain age, and as a result they make their resumes stronger in all future periods. This is a general property that will be satisfied in other information structures as discussed in Appendix B.2.3.

[41]Moreover, if firms believe there is more human capital depreciation than there actually is, firms may mistakenly discount too much experiences in the past as if workers human capital had depreciated.

These three observations are summarized in Proposition 7, below.

Proposition 7. *For any distribution of preferences and productivities, and for any ϕ, where $\mathtt{I} = \int_0^a \phi(\tilde{a}, a)\tilde{h}(\tilde{a})d\tilde{a}$, with $\phi > 0$ and $\tilde{h} \geq 0$ bounded and continuous, and provided that the workers who are more productive are more willing to supply deliverables:*

1. *All workers start with an empty resume and earn the same initial salaries;*

2. *At the peak of the career of each worker, salaries are higher than productivities;*

3. *There is a positive signaling return to experience.*

B.2 Model Extensions

To add realism to an otherwise stylized model, several extensions to the basic model are presented. These extensions include human capital accumulation, and richer signal structures. Key insights from the generalized optimal taxation formulas and their empirical implications will hold with some caveats in those extensions.

The first extension allows human capital accumulation in the form of learning-by-doing as in Arrow (1962), and relaxes the assumption that workers have constant productivities over their lifetimes, which is an evidently implausible assumption. It is reasonable to expect that at least some part of the return to experience observed in the data is due to increases in productivity due to on-the-job learning, or training efforts. However, the extended model shows that this assumption is to some extent innocuous. Although it complicates the relationship between the observed return to experience, the degree of information asymmetry in the market and the rate of human capital accumulation, the same optimal tax formula applies when on-the-job learning is costless.

The second extension allows firms to see additional signals that the government does not see. For example, from the point of view of the employers it may be clear that some workers are on different career tracks, and that those workers can hardly change that. But for the government, it may be hard to distinguish between them, or it may be hard to codify those distinctions into the tax system in a way that cannot be manipulated. In this case, optimal taxes are described by a weighted version of the basic taxation formula, where the weights are given by the sensitivity of the different post-tax retention functions to changes in marginal taxes. This modification of the optimal tax formulas follows from the fact that there are multiple career tracks and there are multiple pre-tax salary functions, while there is a single nonlinear taxation instrument the government can use. Considering a variation in the income retention schedule and, then, tracking how this variation affects the post-tax salaries of different careers in response to it results in the weighted version of the basic optimal tax formula. The effects of changes in marginal taxes on post-tax salaries are attenuated by changes in pre-tax salaries whenever there are information asymmetries. For this reason, taking into account these different career tracks may attenuate the magnitude of the corrective component of taxes.

The third extension, similarly, considers the possibility that full history of completion of tasks may be observed by firms, while the government can see the history of earnings. Considering that possibility requires simultaneously introducing richer heterogeneity in preferences, so that firms can form meaningful expectations of productivities conditional on different histories of completion of deliverables. It is shown that the same general principles apply and a production efficiency argument also holds.

B.2.1 On-the-job Learning

The assumption that workers have a constant productivity over their lifetimes is, of course, extreme. It is reasonable to expect that at least some part of the return to experience observed in the data is due to increases in productivity due to on-the-job learning, or training efforts. A simple way to accommodate these concerns is to allow for productivities to depend not only on the types of workers but also on experience itself, that is $v = v(\theta, h)$. Indeed, one of the key reasons why employers may focus on experience as a signal for the productivity of workers is exactly because accumulating experience may directly increase the productivity of workers.

Revisiting the example from Section 3.1, a simple way to enrich its setup is to assume that productivities increase proportionally with experience, that is, productivities are $v(\theta, h) = \tilde{v}(\theta)h^{\beta}$. Then, because wages are the expectation of productivities conditional on h, $w(h) = \mathbb{E}[v(\tilde{\theta}, h)|\tilde{\theta} \geq \theta(h), h] = h^{\beta}\mathbb{E}[\tilde{v}(\tilde{\theta})|\tilde{\theta} \geq \theta(h)]$, and the return to experience would have two components, a signaling and a human capital accumulation component. Salaries still satisfy a log-linear relationship, with a coefficient that is the sum of the signaling $\frac{\delta}{1-\delta+\epsilon}$ and the human capital component β:

$$log(w) = \Big(\frac{\delta}{1-\delta+\epsilon} + \beta\Big) \cdot log(h) + \Big(\frac{(1-\delta)(1+\epsilon)}{(1-\delta)(1+\epsilon) - \delta\epsilon}\Big) \cdot log\Big(\frac{\alpha}{\alpha-\delta}\Big) \quad (69)$$

More generally, while human capital accumulation very much changes the meaning of the return to experience, the next proposition, focusing on the simpler case where the resume is defined as the cumulative discounted sum of deliverables, shows that the necessary condition for the optimality of taxes from 11 is unchanged.

Proposition 8. *Suppose productivity depends on experience and the unobserved types of workers. The total contribution to output from a worker is equal to* $\int_0^{h(\tilde{R},\theta)} v(\theta, \tilde{h})d\tilde{h}$*, and, thus, we can write the planner's problem as:*

$$\max_{\tilde{R}(h)} \mathbb{E}[W(V(\tilde{R};\theta), \theta)] \ s.t. \ \mathbb{E}\Big[\int_0^{h(\tilde{R},\theta)} (v(\theta, \tilde{h}) - \tilde{r}_h(\tilde{h}))d\tilde{h} - I\Big] \geq 0 \quad (70)$$

Moreover, a necessary condition for a tax schedule to be optimal is given by the following formula (which is analogous to equation 11):

$$\left(\frac{\chi(y)-r(y)}{r(y)}\right)\epsilon_r^c(y)g(y)y=\int_y^\infty\left(1-\lambda(\tilde{y})\right)g(\tilde{y})d\tilde{y}+\int_y^\infty\left(\frac{\chi(\tilde{y})-r(\tilde{y})}{r(\tilde{y})}\right)\eta_I(\tilde{y})g(\tilde{y})d\tilde{y},\ (71)$$

where $\chi(y)=\frac{v(y)}{y'(h(y))}$, and $v(y)=v(\theta(y),h(\theta(y)))$ is the productivity at retirement of the worker with lifetime income y. The first order condition from Problem 70 is identical to the first order condition from Problem 26, where there was no on the job learning. Therefore the resulting necessary condition for taxes, Equation 71 is identical to Equation 11. While the formula looks is the same, there is a subtle but important distinction. We cannot use estimates of productivity of workers far from retirement to infer their productivity at retirement, as these two quantities can be significantly different. Second, we cannot infer the degree of informational asymmetry by only looking at the return to experience. For these reasons, the empirical strategies aimed at estimating $\chi(y)$ applied in Section 5 are designed to be robust to these concerns, and to aim precisely at disentangling the return to experience coming from human capital accumulation from the return to experience coming from employers learning about productivities through job histories. More generally, the left hand side of Equation 14 can be reinterpreted to accomodate human capital accumulation. That is, we can think of $v(\theta)$ as how much one extra deliverable adds to the product of the economy not only now but also in the future, increasing the productivity of future effort. Under that view, the Pigouvian component of taxes should guarantee that the present value of benefits the worker receives from generating one extra unit of the deliverables should be equal to the present value of contributions to output – interpreted broadly to include future productivity increases.

B.2.2 Richer Signal Structure, Exogenous Signals

The baseline version of the model features a very simple signal structure. This simplicity allows us to clearly understand how incentives and the distribution of income interact when there are career concerns. However, it leaves out important elements of real labor markets, such as richer signals that firms can extract from workers. For instance, firms may clearly recognize that some workers are on different career tracks, and for those workers it may be very difficult to change career tracks. For the government, it may be harder to distinguish these workers, and even harder to create taxes that are specific to each career track. This possibility can be accommodated by introducing exogenous signals that only the firms and not the government sees.

If firms see additional exogenous signals the government does not see, then we cannot apply Proposition 2. The reason behind this is that there are multiple pre-tax salary functions, a different career path for each signal realization. However, we can still consider a variation in the income retention schedule and track how this variation affects the salaries of different careers in response to it. The following proposition uses this idea to derive comparable tax formulas in this more complex environment, but focusing on the simpler case where the resumes are defined as the cumulative discounted sum of deliverables.

Proposition 9. *If a tax schedule is optimal then it needs to satisfy the following relationship:*

$$\mathbb{E}_z\left[\int_y^\infty \left(\frac{\chi(\tilde{y},z)-r_{\tilde{y}}}{r_{\tilde{y}}}\right)\epsilon^c_{r_{\tilde{y}}}(\tilde{y},z)g(\tilde{y}|z)\tilde{y}\frac{dr_{h(\tilde{y};z)}}{dr_y}d\tilde{y}\right] = \mathbb{E}_z\left[\int_y^\infty\int_{\tilde{y}}^\infty \left(1-\lambda(\tilde{\tilde{y}};z)\right)g(\tilde{\tilde{y}}|z)\cdot\frac{dr_{h(\tilde{y};z)}}{dr_y}d\tilde{\tilde{y}}d\tilde{y}\right]$$
$$+\mathbb{E}_z\left[\int_y^\infty\int_{\tilde{y}}^\infty \left(\frac{\chi(\tilde{\tilde{y}},z)-r_{\tilde{\tilde{y}}}}{r_{\tilde{\tilde{y}}}}\right)\eta^h_I(\tilde{\tilde{y}})g(\tilde{\tilde{y}}|z)\cdot\frac{dr_{h(\tilde{y};z)}}{dr_y}d\tilde{\tilde{y}}d\tilde{y}\right] \quad (72)$$

where $\chi(\tilde{y},z)=\frac{v(h(\tilde{y},z);z)}{y'(h(\tilde{y},z);z)}$, *and* $\frac{dr_{h(\tilde{y};z)}}{dr_y}$ *is the response of post-tax salaries of someone who initially earns income* y*, and who gets the signal* z*.*

In this case the optimal policy is described by a weighted version of the standard first order condition, where weights are given by how much post-tax wages change when income taxes change at different career paths. Taking into account this heterogeneity across career tracks may attenuate the size of the corrective component of taxes. For example, if there are no income effects, then $\frac{dr_{h(\tilde{y};z)}}{dr_y}=0$ for $\tilde{y}\neq y$, and the formula above reduces to:

$$\mathbb{E}_z\left[\left(\frac{\chi(y,z)-r_y}{r_y}\right)\epsilon^c_{r_y}(y,z)g(y|z)y\frac{dr_{h(y;z)}}{dr_y}\right] = \mathbb{E}_z\left[\frac{dr_{h(y;z)}}{dr_y}\int_y^\infty \left(1-\lambda(\tilde{y};z)\right)g(\tilde{y}|z)\cdot d\tilde{y}\right] \quad (73)$$

For any signal realization z, the weights $\frac{dr_{h(y;z)}}{dr_y}$ are equal to one if there is no information asymmetry and are less than one if there is any informational asymmetry.[42] In this sense, relative to the basic equation 11, holding fixed the other sufficient statistics, taking into account this source of heterogeneity would call for relatively lower marginal taxes, attenuating the intensity with which informational asymmetries push toward higher marginal taxes.

Firms See Additional Signals (Proof of Proposition 9)

Proof. Consider the problem:

$$\max_{\tilde{R}} \mathbb{E}[W(V(\tilde{R},\theta),\theta)] \ s.t.\ \mathbb{E}[v(\theta)(h(\theta))-\tilde{R}(h(\theta))]\geq 0 \quad (74)$$

Considering a small variation on marginal retention rates as a function of earnings, and noticing that this variation translates into a $\frac{dr_h}{dr_y}$ variation in the marginal retention as a function of effort

[42]For more details on how changes in salaries and the degree of informational asymmetry χ are related see Section 5.2.

(as in the Proof of Proposition 2, Appendix A.4), we can write a weighted version of Equation 29.

$$\mathbb{E}\left[\lambda(\theta)\frac{dV(r,\theta)}{dr_h}\cdot\frac{dr_h}{dr_y}\right] = -\mu\mathbb{E}\left[v(\theta)\frac{dh(\theta)}{dr_h}\cdot\frac{dr_h}{dr_y} - \frac{dh(\theta)}{dr_h}\cdot\frac{dr_h}{dr_y}r_h(h(\theta),z(\theta)) - 1(y(h(\theta),z(\theta))\geq y)\cdot\frac{dr_h}{dr_y}\right] \tag{75}$$

Taking analogous steps to those in the Proof of Proposition 2, we conclude that:

$$\begin{aligned}\mathbb{E}_z\left[\int_y^\infty\left(\frac{\chi(\tilde{y},z)-r_{\tilde{y}}}{r_{\tilde{y}}}\right)\epsilon^c_{r_{\tilde{y}}}(\tilde{y},z)g(\tilde{y},z)\tilde{y}\frac{dr_{h(\tilde{y};z)}}{dr_y}d\tilde{y}\right] &= \mathbb{E}_z\left[\int_y^\infty\int_{\tilde{y}}^\infty\left(1-\lambda(\tilde{\tilde{y}};z)\right)g(\tilde{\tilde{y}}|z)\cdot\frac{dr_{h(\tilde{y};z)}}{dr_y}d\tilde{\tilde{y}}d\tilde{y}\right] \\ &+\mathbb{E}_z\left[\int_y^\infty\int_{\tilde{y}}^\infty\left(\frac{\chi(\tilde{\tilde{y}},z)-r_{\tilde{\tilde{y}}}}{R(\tilde{\tilde{y}})}\right)\eta^h_I(\tilde{\tilde{y}})g(\tilde{\tilde{y}}|z)\cdot\frac{dr_{h(\tilde{y};z)}}{dr_y}d\tilde{\tilde{y}}d\tilde{y}\right]\end{aligned} \tag{76}$$

□

B.2.3 Conditioning on the Full History of Deliverables

In addition to the single scalar representation of resumes discussed throughout the paper, we consider the case of a more general information structure in which firms observe the full history of deliverables, and the detailed timing of their execution. That is, for a worker of age a, the firm will see the history of labor supply decisions from the time the worker was born up to their current age a, which is denoted $\tilde{h}^a$.[43]

We assume in this section that preferences now take the form $\mathrm{U}(C,\tilde{h},\theta)$, and that θ is high-dimensional. To talk meaningfully about expectations of productivities conditional on some history $\tilde{h}^a$, we need some workers to be willing to provide this rich set of histories, which would not be possible for preferences of the form $U(C,H(\tilde{h})),\theta)$, or with a low dimensional type space. To keep this section simple, we additionally assume there is a single consumption good C.[44] We retain the assumption that more productive types are more willing to provide the deliverables.[45]

Under those assumptions, we will show that optimal taxes can also be thought of as the composition of i) corrective taxes that guarantee that, for each additional unit of labor, workers receive lifetime benefits equivalent to their contribution to output, and ii) redistributive taxes, which, in this case, do not necessarily take the simple Mirrleesian form of Section 4.

Toward showing this result, analogously to what we did before, first we establish that if the planner could choose the allocation, while being restricted to the set of incentive-compatible allocations, any optimal allocation would lie at the frontier of the production possibilities set (Lemma 6 in Appendix B.2.3). Then, we show that the planner can use

[43]In this case, workers of different ages are never pooled together, as any history that ends up at age a is different than a history that ends at age a', when $a \neq a'$.

[44]This is without loss, since discount rates q are assumed to be exogenous.

[45]Which in this case can be stated as: if $v(\theta) > v(\theta')$ then for any $\tilde{h}$, C, $MRS_{C,\tilde{h}(a)}(C,\tilde{h},\theta) < MRS_{C,\tilde{h}(a)}(C,\tilde{h},\theta')$, where $MRS_{C,\tilde{h}(a)}(C,\tilde{h},\theta) = -\frac{U_{\tilde{h}(a)}(C,\tilde{h},\theta)}{U_C(C,\tilde{h},\theta)}$.

Pigouvian taxes to achieve the frontier of the production efficient set of allocations. This will be the case because the planner will be able to infer labor supply choices from the history of earnings, that is, labor supply $\tilde{h}$ will map to earnings $\tilde{y}$ one-to-one, as in Lemma 2 and Lemma 5.

To show that the planner can infer labor supply choices from the history of earnings, an important intermediate result is that there is a positive signaling return to experience. This result is interesting on its own, and it shows that the career concerns logic carries through this more general environment (see Lemma 7 in Appendix B.2.3, which is the analogous counterpart to Lemma 4) . This signaling return to experience is driven by the fact that higher productivity types are those who are willing to provide labor supply paths with a larger amount of deliverables. That property is used to show that the planner can infer labor supply decisions from earnings histories (see Lemma 8 in Appendix B.2.3), which in turn is crucial for the ability of the planner to implement allocations at the frontier of production possibilities set. That is, analogously to Lemma 7 and Lemma 4, from the Section 4.2 and Appendix A.5, Lemma 8 in this section allows the planner to decentralize feasible, incentive compatible allocations that lie at the efficient possibilities frontier of the economy with the use of taxes on earnings. In doing that, Pigouvian taxes play an important role, as stated in the following Proposition.

Proposition 10. *The planner can guarantee that the allocation would lie at the frontier of the production possibilities set by using Pigouvian taxes.*

These Pigouvian taxes take the same general form as in Proposition 3 . Thus, although this economy may look quite complicated, the same principles of tax design can be applied. There is a caveat, though. Because we have unrestricted preferences and multiple goods, now the design of optimal redistributive taxes, after correcting for the Pigouvian distortions, is more complicated, and without further normative assumptions, we cannot point to lifetime income taxation as the preferred form of redistribution. From an implementation perspective, analogously to the cases analyzed in Section 4, the structure of taxes can be thought of as a composition of two layers of taxes. First, Pigouvian taxes correct for the distortions. Second, on top of these taxes, another layer of taxes is imposed to take care of redistribution.

Proof of Proposition 10

Lemma 6. *If the planner could choose the allocation, while being restricted to the set of incentive compatible allocations, any optimal allocation would lie at the frontier of the production possibilities set.*

Proof. Follows from analogous arguments from the production efficiency theorem of

Diamond and Mirrlees (1971). That is, consider the problem

$$\max_{C_\theta,\tilde{h}_\theta} \mathbb{E}[W(\mathrm{U}(C_\theta,\tilde{h}_\theta,\theta),\theta)] \ st\ \mathbb{E}[v(\theta)\int_0^1 q(a)\tilde{h}_\theta(a)da - C_\theta] \geq 0 \tag{77}$$

$$U(C_\theta,\tilde{h}_\theta,\theta) \geq U(C_{\theta'},\tilde{h}_{\theta'},\theta)\ \forall\theta,\theta'$$

By the taxation principle, we can incorporate the incentive compatibility constraints into the indirect utility function of the workers and solve the equivalent problem:

$$\max_{R(\tilde{h}_\theta)} \mathbb{E}[W(\mathrm{V}(R,\theta))] \ st\ \mathbb{E}[v(\theta)\int_0^1 q(a)\tilde{h}_\theta(a;R)da - R(\tilde{h}_\theta(\cdot;R))] \geq 0 \tag{78}$$

Now, toward a contradiction, suppose production takes place at the interior of the production possibility frontier. Then, we can increase $R(\tilde{h}_\theta)$ uniformly (as lowering the price of the consumption good). Because the indirect utility function is increasing in R, every worker would be better off, and welfare would be higher. This is feasible, because under the assumption that labor supply decisions are continuous in a uniform increase in R, there is a small enough increase in $R(\tilde{h}_\theta(\cdot))$ that keeps the allocation inside the production possibilities set. □

The second result is that the planner can use Pigouvian taxes to achieve the frontier of the production efficient set of allocations, because labor supply $\tilde{h}$ will map to earnings $\tilde{y}$ one-to-one, as in Lemma 2 and Lemma 5.

Lemma 7. *There is a positive return to experience. That is, in any optimal allocation, salaries $w(\tilde{h}^a) = \mathbb{E}[v(\theta)|\tilde{h}^a]$ are increasing in labor supply choices $\tilde{h}(\tilde{a})$, where $\tilde{a} \leq a$.*

Proof. This is an immediate consequence of the assumption that the more productive types are more willing to provide the deliverables, and that the type space is rich enough so that for any path $\tilde{h}$, expectations are well-defined. □

This Lemma, analogously to 4, establishes that there is positive return to experience.

Lemma 8. *The planner can infer labor supply choices from earnings. That is, no two continuous $\tilde{h}$ map to the same $\tilde{y}$.*

Proof. Remember that $\tilde{y}(a,\tilde{h}^a) = \tilde{h}(a)\cdot w(\tilde{h}^a) = \tilde{h}(a)\cdot\mathbb{E}[v(\theta)|\tilde{h}^a]$. Consider the first non-zero measure interval where $\tilde{h}$ and $\tilde{h}'$ differ, and without loss consider a ball $(\underline{a},\bar{a})$ where $\tilde{h}(a) > \tilde{h}'(a)$. If salaries were the same for both functions, $\tilde{y}(a,\tilde{h}^a) > \tilde{y}(a,\tilde{h}'^a)$, and the proof would be complete. However, by the previous Lemma, salaries are increasing as a function of $\tilde{h}(a)$, for all a; thus indeed we have that $\tilde{y}(a,\tilde{h}^a) > \tilde{y}(a,\tilde{h}^a)$. □

Those results imply Pigouvian taxes should play an important role, as stated in the following Proposition.

Proposition. *The planner can guarantee that the allocation would lie at the frontier of the production possibilities set by using Pigouvian taxes.*

Proof. Consider the problem 78. The solution to this problem results in a wage schedule, $R(\tilde{h})$, which is a functional of labor supply decisions. This wage schedule, by the previous Lemma, can be written as a functional of the history of earnings $\tilde{y}$, $R(\tilde{y})$, so it can be implemented with history-dependent earnings taxes. If there is separation, so that workers of different productivites supply different labor supply histories, we can write and think of that solution as $\tilde{R}(\tilde{h}) = R_m(\tilde{R}_p(\tilde{h}))$, where $\frac{\partial R_p(\tilde{h})}{\partial \tilde{h}(a)} = v(\tilde{h})q(a)$.[46] $\square$

B.3 Equilibrium Existence Without Taxes

Proposition 11. *Assume that $v(\theta)$ smoothly increasing in types, $MRS(c,h,\theta) \equiv -\frac{U_h(c,h,\theta)}{U_c(c,h,\theta)}$ smoothly decreasing in types, consumption and labor, and that the distribution of types is continuous. Assume as well that $MRS_c + MRS_h w$, where $w(\theta) \equiv \mathbb{E}[v(\tilde{\theta})|\tilde{\theta} \geq \theta]$, is bounded away from zero and that MRS_θ, MRS_h and MRS_c are uniformly Lipschitz continuous in c and h. Then, there exists an equilibrium.*

Proof. Consider the allocation $c(\theta), h(\theta)$, where $c(\theta)$ and $h(\theta)$ is the solution to the following system of differential equations

$$\frac{c'(\theta)}{h'(\theta)} - MRS(c(\theta), h(\theta), \theta) = 0 \tag{79}$$

$$\mathbb{E}[v(\tilde{\theta})|\tilde{\theta} \geq \theta] = \frac{c'(\theta)}{h'(\theta)} \tag{80}$$

with boundary conditions:

$$\frac{c(\underline{\theta})}{h(\underline{\theta})} - MRS(c(\underline{\theta}), h(\underline{\theta}), \theta) = 0 \tag{81}$$

$$c(\underline{\theta}) = h(\underline{\theta}) \cdot \mathbb{E}[v(\theta)] \tag{82}$$

We claim that this allocation satisfies incentive compatibility and is feasible. The first equation is the local incentive compatibility constraint. $c(\theta)$ and $h(\theta)$ are increasing in θ because $\mathbb{E}[v(\tilde{\theta})|\tilde{\theta} \geq \theta]$ is increasing in θ and MRS is decreasing in θ, and increasing in c and h. Because of single-crossing, the local incentive compatibility constraints and monotonicity together imply that incentive constraints are globally satisfied. Note that the feasibility constraint $\int (c(\theta) - v(\theta)h(\theta)))f(\theta)d\theta = 0$ is automatically satisfied given the first two equations, as workers get paid what they produce in expectation, firms are offering

[46] If there is not separation, then we can still think of the solution as $\tilde{R}(\tilde{h}) = R_m(\tilde{R}_p(\tilde{h}(\cdot;\theta)), \theta)$, where $\frac{\partial R_p(\tilde{h})}{\partial \tilde{h}(a)} = v(\theta)q(a)$, which can in a weaker sense also be interpreted as composition of two layers of taxes, one of those being Pigouvian.

salaries $\mathbb{E}[v(\tilde{\theta})|\tilde{\theta} \geq \theta]$ and making zero profits. The existence of a solution is guaranteed by the standard existence theorems for initial value problems, given the regularity assumptions on MRS, v and the distribution of types (see, for example, Coddington et al. (1956)).[47]

$\square$

B.4 First Best with Pigouvian Taxes

Proposition 12. *Assume that $v(\theta)$ smoothly increasing in types, $MRS(c,h,\theta) \equiv -\frac{U_h(c,h,\theta)}{U_c(c,h,\theta)}$ smoothly decreasing in types, consumption and labor, and that the distribution of types is continuous. Assume as well that $MRS_c + MRS_h v$ is bounded away from zero and that MRS_θ, MRS_h and MRS_c are uniformly Lipschitz continuous in c and h. Then, a first best allocation can be achieved with $r(y) = \chi(y)$.*

Proof. Consider the direct mechanism that offers the allocation $c(\theta), h(\theta)$, where $c(\theta)$ and $h(\theta)$ is the solution to the following system of equations.

$$\frac{c'(\theta)}{h'(\theta)} - MRS(c(\theta), h(\theta), \theta) = 0 \tag{85}$$

$$v(\theta) = \frac{c'(\theta)}{h'(\theta)} \tag{86}$$

with boundary conditions:

$$\int (c(\theta) - v(\theta)h(\theta)))f(\theta)d\theta = 0 \tag{87}$$

$$MRS(c(\underline{\theta}), h(\underline{\theta}), \theta) = v(\underline{\theta}) \tag{88}$$

We claim that this allocation satisfies incentive compatibility constraints and it is feasible. The first equation is the local incentive compatibility constraint. $c(\theta)$ and $h(\theta)$ are increasing in θ because $v(\theta)$ is increasing in θ and MRS is decreasing in θ, and increasing in c and h. Because of single-crossing, the local incentive compatibility constraints together with monotonicity implies that global incentive compatibility constraints are satisfied. The third equation is the feasibility constraint. Notice as well that $\int_{\underline{\theta}}^{\theta} v(\tilde{\theta})h'(\tilde{\theta})d\tilde{\theta} < v(\theta)\int_{\underline{\theta}}^{\theta} h'(\tilde{\theta})d\tilde{\theta} \leq v(\theta)h(\theta)$, thus $c(\underline{\theta}) > 0$, that is, higher types workers in this allocation

[47]Notice we can rearrange the equations above (combine the first and second equations, take a derivative with respect to θ, and replace $h'(\theta) = c'(\theta)\mathbb{E}[v(\tilde{\theta})|\tilde{\theta} \geq \theta]$) to write our system of differential equations as :

$$c'(\theta) = \frac{\frac{d\mathbb{E}[v(\tilde{\theta})|\tilde{\theta}\geq\theta]}{d\theta} - MRS_\theta}{MRS_c + MRS_h\mathbb{E}[v(\tilde{\theta})|\tilde{\theta} \geq \theta]} \tag{83}$$

$$h'(\theta) = \frac{\left(\frac{d\mathbb{E}[v(\tilde{\theta})|\tilde{\theta}\geq\theta]}{d\theta} - MRS_\theta\right)\mathbb{E}[v(\tilde{\theta})|\tilde{\theta} \geq \theta]}{MRS_c + MRS_h\mathbb{E}[v(\tilde{\theta})|\tilde{\theta} \geq \theta]} \tag{84}$$

Given our regularity assumptions right-hand side of Equations 83 and 84 are continuous in θ and uniformly Lipschitz continuous in c and h, therefore, there exists a solution.

subsidize lower type workers and consumption is positive everywhere. Further, notice that in this allocation $r(y) = \chi(y)$. The existence of a solution is guaranteed by standard existence theorems for the solution of ordinary differential equations, given the regularity assumptions on MRS, v and the distribution of types (see, for example, Coddington et al. (1956)).[48]

Finally, we show that it is a first best allocation: it corresponds to an allocation where workers face linear budgets and get paid their marginal products while receiving transfers $I(\theta) = c(\theta) - w(\theta)h(\theta)$. □

B.5 Comparative statics on taxes

To understand how the Mirrleesian component and total marginal taxes would be affected, it is useful to write optimal taxes in terms of types θ, as in the following proposition.

Proposition 13. *Optimal taxes as a function of types θ must satisfy the following equations:*

$$r_y(\theta) = r_m(\theta) \cdot \chi(\theta) \tag{91}$$

$$\frac{1 - r_m(\theta)}{r_m(\theta)} f(\theta) \Big(- \frac{\partial \log MRS}{\partial \theta} \Big)^{-1} = \int_{\theta}^{\infty} (1 - \hat{\lambda}(\tilde{\theta})) f(\tilde{\theta}) d\tilde{\theta} + \int_{\theta}^{\infty} \Big(\frac{1 - r_m(\tilde{\theta})}{r_m(\tilde{\theta})} \Big) \eta(\tilde{\theta}) f(\tilde{\theta}) d\theta \tag{92}$$

Changes in the degree of informational asymmetry affect at least two key ingredients in the formula above χ and $\frac{\partial \log MRS}{\partial \theta}$, and each of them, respectively, affects directly the Mirrleesian and the Pigouvian components of taxes. They also operate very differently. The Pigouvian component of taxes for a type θ is affected by the changes in the unobservable component of the productivity of everyone that ends up with a lifetime income higher than $y(\theta)$. As the type of labor these people supply become more easily measurable, type

[48] Notice we can rearrange the equations above (combine the first and second equations, take a derivative with respect to θ, and replace $h'(\theta) = c'(\theta)v(\theta)$) to write our system of differential equations as :

$$c'(\theta) = \frac{v'(\theta) - MRS_\theta}{MRS_c + MRS_h v(\theta)} \tag{89}$$

$$h'(\theta) = \frac{\Big(v'(\theta) - MRS_\theta\Big)v(\theta)}{MRS_c + MRS_h v(\theta)} \tag{90}$$

Given our regularity assumptions, the right-hand side of Equations 89 and 90 are continuous in θ and uniformly Lipschitz continuous in c and h. Therefore, given initial values, there exists a solution. The solution is also continuous in the initial values. Thus, we can define the initial value implictly by Equations 87 and 88, since the resource constraint could be made slack or violated by changing the initial values of c_0, while holding that $MRS(c_0, h_0, \theta) = v_0$. By making the lowest-type workers receive the product they generate (i.e. setting $c(\underline{\theta}) = h(\underline{\theta})v(\underline{\theta})$), we would conclude that the resource constraint would be slack. Conversely, taking the limit where the highest-type workers receive the product they generate, we would conclude that the resource constraint is violated. Thus, there exists an initial value that guarantees that the resource constraint is satisfied with equality.

θ receives smaller implicit subsidies from higher productivity workers, and $\chi(\theta)$ increases, approaching one. The Mirrleesian component, on the other hand, depends on how type θ is more or less willing to provide more deliverables relative to their local neighboring types. Thus, for any given θ, one can imagine a decrease in informational asymmetries that can have impacts on the Mirrleesian or Pigouvian component of taxes of arbitrarily different magnitudes. That is, the net effect on total marginal taxes, taking into account both the Mirrleesian and Pigouvian components, is in general ambiguous.

A simple example shows that we may expect the Pigouvian component to dominate over the Mirrleesian component in certain circumstances. Assuming there are no income effects, and that preferences take the simple form as in the example from Section 3.1, and types are Pareto distributed, we can solve for optimal marginal retention as a function of types as:

$$\chi = \frac{\alpha-\delta}{\alpha}, \quad \frac{1-r_m}{r_m} = \frac{1-\bar{\lambda}}{\frac{\epsilon}{1+\epsilon}\frac{\alpha}{1-\delta}} \quad \implies r = r_m \cdot \chi = \frac{\frac{\epsilon}{1+\epsilon}\frac{\alpha-\delta}{1-\delta}}{(1-\bar{\lambda}) + \frac{\epsilon}{1+\epsilon}\frac{\alpha}{1-\delta}} \tag{93}$$

where $\bar{\lambda}(\theta) = \mathbb{E}[\hat{\lambda}(\tilde{\theta})|\tilde{\theta} \geq \theta]$ denotes the marginal value of a one dollar transfer to the types above θ, which is assumed to be constant in θ starting from some level $\hat{\theta}$. χ is decreasing in the degree of informational asymmetry; thus corrective taxes increase as informational asymmetries increase. r_m is increasing in the degree of informational asymmetry, and therefore the redistributive component of taxes decrease. Provided that $\frac{\alpha}{1+\epsilon} \geq 1$ (which incidentally guarantees that output is finite), holding fixed this marginal value of transfers $\bar{\lambda}$, marginal retention r decreases with the degree of informational asymmetry δ, that is, marginal taxes increase as the degree of informational asymmetry increases. In this case, thus, the Pigouvian component dominates over the Mirrleesian component, calling for higher marginal taxes as the degree of informational asymmetry increases.

Proof of Proposition 13

Proof. The proof adapts the derivation of optimal taxation formulas from Scheuer and Werning (2017) to our enviroment. Let's set up the planner's problem as:

$$\begin{aligned} \underset{u,h}{max} \ & \int W(u(\theta),\theta)f(\theta)d\theta \\ s.t. \ & u'(\theta) = U_\theta(e(u(\theta),h(\theta),\theta),h(\theta),\theta) \\ & \int (e(u(\theta),h(\theta),\theta) - v(\theta)h(\theta))f(\theta)d\theta \leq 0 \end{aligned} \tag{94}$$

We can write the Lagrangian, where for conciseness, we drop the explicit dependence on θ

$$\underset{u,h}{max} \int W(u) \ f + \mu(u' - U_\theta) - \kappa(e - vh)f \ d\theta \tag{95}$$

Integrate by parts

$$\underset{u,h}{max} \int W(u)\ f - \mu' u - \mu U_\theta - \kappa(e - vh) f\ d\theta + u\mu|_{\underline{\theta}}^{\overline{\theta}} \tag{96}$$

Take FOC's for $u(\theta)$, and $h(\theta)$, denote $\lambda(\theta) = W'(u(\theta), \theta)$

$$\lambda f - \mu' - \mu U_{\theta,c} e_u - \kappa e_u f = 0 \tag{97}$$

$$\mu(U_{\theta,c} e_h + U_{\theta,h}) - k e_h f + kvf = 0 \tag{98}$$

Replace $e_u = U_c^{-1}$ on Equation 97

$$\lambda U_c f - \mu' U_c - \mu U_{\theta,c} - f\kappa = 0 \tag{99}$$

Define $\hat{\mu} = \mu U_c/\kappa$ and $\hat{\lambda} = \lambda U_c/\kappa$, and rearrange, denoting $MRS_c = \frac{dMRS(c,h,\theta)}{dc}$

$$\hat{\lambda} f - \hat{\mu}' - \hat{\mu} MRS_c h' = f \tag{100}$$

Rearrange Equation 98

$$-\hat{\mu} = \frac{f\ \frac{v-r}{r}}{\frac{\partial \log MRS}{\partial \theta}} \tag{101}$$

Differentiate Equation 101:

$$-\hat{\mu}' = \frac{f'\ \frac{v-r}{r} + f\ \frac{d}{d\theta}(\frac{v-r}{r})}{\frac{\partial \log MRS}{\partial \theta}} - \frac{\frac{\partial^2 \log MRS}{\partial \theta^2} f\ \frac{v-r}{r}}{\frac{\partial \log MRS}{\partial \theta}^2} \tag{102}$$

Plugging back in the Equation 100:

$$\hat{\lambda} f + \frac{f'\ \frac{v-r}{r} + f\ \frac{d}{d\theta}(\frac{v-r}{r})}{\frac{\partial \log MRS}{\partial \theta}} - \frac{\frac{\partial^2 \log MRS}{\partial \theta^2} f\ \frac{v-r}{r}}{\frac{\partial \log MRS}{\partial \theta}^2} + \frac{f\ \frac{v-r}{r}}{\frac{\partial \log MRS}{\partial \theta}} MRS_c h' = f \tag{103}$$

Rearrange Equation 103

$$\frac{v-r}{r}\Big(f'\Big(\frac{\partial \log MRS}{\partial \theta}\Big)^{-1} - f\frac{\frac{\partial^2 \log MRS}{\partial \theta^2}}{\frac{\partial \log MRS}{\partial \theta}^2}\Big) + f\Big(\frac{\partial \log MRS}{\partial \theta}\Big)^{-1} \frac{d}{d\theta}\Big(\frac{v-r}{r}\Big) = (1 - \hat{\lambda}) f - \Big(\frac{v-r}{r}\Big) f \Big(\frac{\partial \log MRS}{\partial \theta}\Big)^{-1} MRS_c h' \tag{104}$$

Integrate both sides with respect to θ

$$\frac{v-r}{r} f\Big(\frac{\partial \log MRS}{\partial \theta}\Big)^{-1}\Big|_\theta^\infty = \int_\theta^\infty (1 - \hat{\lambda}) f d\theta - \int_\theta^\infty \Big(\frac{v-r}{r}\Big) \frac{MRS_c h'}{\frac{\partial \log MRS}{\partial \theta}} f d\theta \tag{105}$$

Notice that: $\eta(\theta) = \frac{dh}{dI} = \frac{MRS_c h'}{\frac{\partial \log MRS}{\partial\theta}}$. Using that $\lim_{\theta\to\infty} \frac{v-r}{r} f\left(\frac{\partial \log MRS}{\partial\theta}\right)^{-1} = 0$:

$$\frac{v-r}{r} f\Big(-\frac{\partial \log MRS}{\partial\theta}\Big)^{-1} = \int_\theta^\infty (1-\hat\lambda) f d\theta + \int_\theta^\infty \Big(\frac{v-r}{r}\Big)\eta f d\theta \tag{106}$$

Finally, notice that $\frac{v-r}{r} = \frac{\chi - r_y}{r_y}$

$$\frac{\chi - r_y}{r_y} f\Big(-\frac{\partial \log MRS}{\partial\theta}\Big)^{-1} = \int_\theta^\infty (1-\hat\lambda) f d\theta + \int_\theta^\infty \Big(\frac{\chi - r_y}{r_y}\Big)\eta f d\theta \tag{107}$$

□

C Empirical Appendix

C.1 Additional Tables and Figures

Table 1: Mental status scores, hourly wages, and participation

mental status scores						
η^m	0.14	-0.82	-1.08	-1.20	-1.30	-1.31
	(0.42)	(0.51)	(0.51)	(0.54)	(0.54)	(0.54)
observations	16027	13187	13187	11030	11030	11030
hourly wages						
ϵ^w	-0.13	-0.061	-0.066	-0.092	-0.16	-0.16
	(0.10)	(0.10)	(0.10)	(0.099)	(0.099)	(0.099)
observations	39179	39179	39179	39179	39179	39179
participation						
η^p	-0.039	0.063	0.044	0.037	0.013	0.010
	(0.056)	(0.058)	(0.056)	(0.062)	(0.062)	(0.062)
observations	72526	72526	72526	61526	61526	61526
year f.e.	no	yes	yes	yes	yes	yes
marital status	no	no	yes	yes	yes	yes
hourly wages	no	no	no	linear	c. spline	l. spline

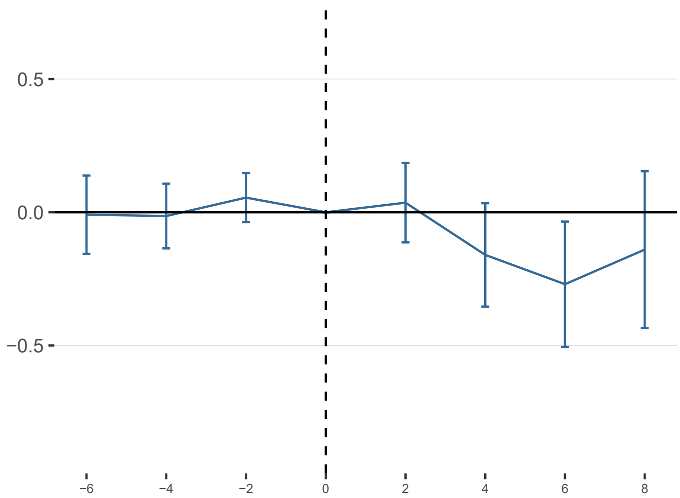
Notes. Robust standard errors in parentheses. Semi-elasticities η^m and elasticities ϵ^w are computed from linear regressions of mental status scores two years in the past, and changes in log hourly wages, respectively, on simulated changes in log marginal retention rates over two years (evaluated at the base year income data), restricting the sample for those who are working in the baseline year and four years ahead. Semi-elasticities η^p are computed from linear regressions of changes of participation over four years on simulated changes in log marginal retention rates. Each column includes different sets of controls: year-fixed effects, marital status dummies, and hourly wages. The fourth column includes log hourly wages. The fifth column includes a 5-piece cubic spline of log hourly wages. The sixth column includes a 10-piece linear spline of log hourly wages.

Table 2: "Rat race" externality estimates $-(1-\chi)$

$-\epsilon_4^w/\eta_4^P$	0.034	-0.014	-0.015	-0.025	-0.12	-0.16
	[-.01, 2.4]	[-2.4, .03]	[-2.7, .02]	[-47, .01]	[-9, -.04]	[-31, -.08]
$-\epsilon_4^w/\eta_2^P$	-0.017	-0.0045	-0.0049	-0.0072	-0.015	-0.015
	[-.22, .02]	[-.03, .01]	[-.03, .01]	[-.04, .01]	[-.12, .00]	[-.13, .01]
year f.e.	no	yes	yes	yes	yes	yes
marital status	no	no	yes	yes	yes	yes
hourly wages	no	no	no	linear	c. spline	l. spline

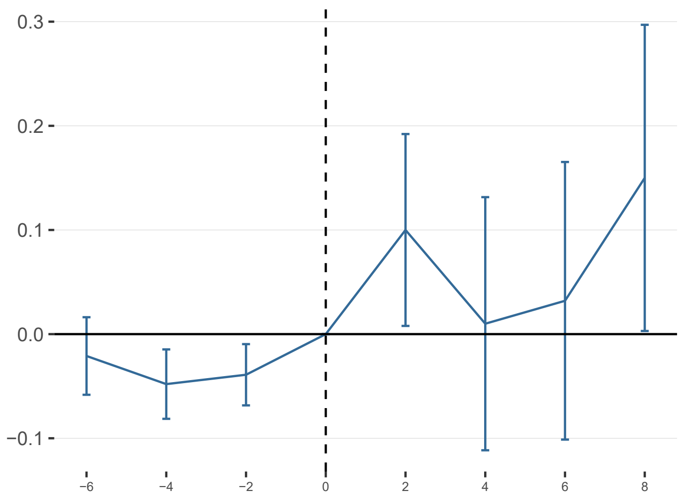
Notes. Bootstrapped bias-corrected confidence intervals in parentheses (with 2000 bootstrap replications). Estimates for externality ($-(1-\chi)$) are obtained by dividing the elasticity of wages by the participation semi-elasticity multiplied by one hundred. Participation semi-elasticities are computed from regressing changes in participation over the next four years (first row, two years in the second row) on changes in log marginal retention rates over two years (evaluated at the base year income data). Each column includes different sets of controls: year fixed effects, marital status dummies, and hourly wages. The fourth column includes log hourly wages. The fifth column includes a 5-piece cubic spline of log hourly wages. The sixth column includes a 10-piece linear spline of log hourly wages.

Figure 5: Elasticity of wages over different horizons



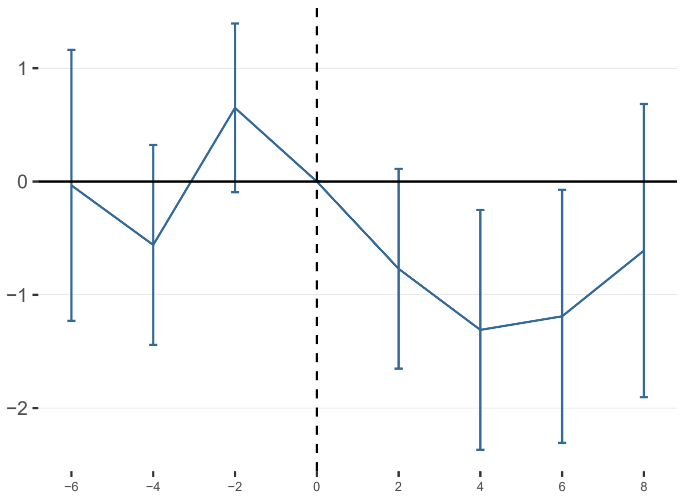
Notes. 95% confidence intervals. Elasticities are computed from linear regressions of changes in log hourly wages over k-years on changes in log marginal retention rates over two years (evaluated at the base year income data). All specifications include year fixed effects, marital status dummies, and a 10-piece linear spline of log hourly wages.

Figure 6: Semi-elasticities of participation over different horizons



Notes. 95% confidence intervals. Semi-elasticities are computed from linear regressions of changes of hours wages over k-years years on changes in log marginal retention rates over two years (evaluated at the base year income data). All specifications include year fixed effects, marital status dummies, and a 10-piece linear spline of log hourly wages.

Figure 7: Semi-elasticities of mental status scores over different horizons

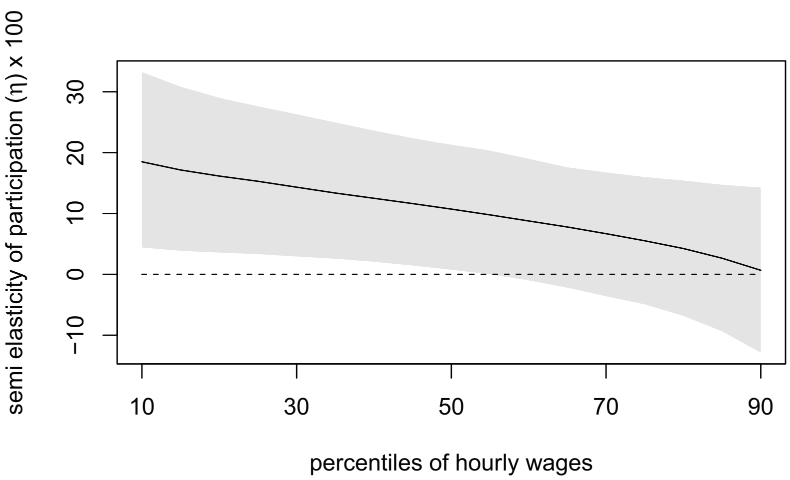


Notes. 95% confidence intervals. For the periods (k) between 2 and 8, semi-elasticities are computed from linear regressions of memory scores two years in the past over years on changes in log marginal retention rates over two years (evaluated at the base year income data), restricting the sample for those who are working in the baseline year and k-years ahead. Similarly, for the periods (k) between -6 to -2, semi-elasticities are computed from linear regressions of memory scores (2+|k|) years in the past over years on changes in log marginal retention rates over two years (evaluated at the base year income data), restricting the sample for those who are working in the baseline year and |k|-years before. All specifications include year fixed effects, marital status dummies, and a 10-piece linear spline of log hourly wages.

Figure 8: Elasticity of wages over different hourly wages percentiles

Notes. 95% bootstrap confidence intervals. Elasticities are computed from local linear regressions of changes in log hourly wages over four years on changes in log marginal retention rates over two years (evaluated at the base year income data), including year fixed effects, marital status dummies, and a 10-piece linear spline of log hourly wages.

Figure 9: Semi-elasticities of participation over different hourly wages percentiles



Notes. 95% bootstrap confidence intervals. Semi-elasticities are computed from linear regressions of changes of hours wages over two years on changes in log marginal retention rates over two years (evaluated at the base year income data), including year fixed effects, marital status dummies, and a 10-piece linear spline of log hourly wages.

C.2 Computing Simulated Marginal Rates and Their Changes

To create a correspondence between HRS variables and TAXSIM 32 variables, we draw extensively from the procedure outlined in RAND's 2014 HRS tax calculations (Pantoja et al., 2018). The correspondence is described in Table 3, and more details are given in the footnotes. Since the procedure is inherently imperfect, after computing tax changes, we restrict the sample to those who have marginal tax rates calculated to be smaller than one hundred percent and have not experienced an unreasonably large simulated change in log marginal retention rates at initial incomes (not larger than 0.3 log points). The restrictions on simulated marginal rates reduce the sample by less than 0.15%. Moreover, we restrict the sample to respondents who, across survey years, have not changed marital status. Further sample restrictions are described in the footnotes of each table and figure.

C.3 Federal and State Variation in Taxes

Over our sample period, federal tax reforms included the Omnibus Budget Reconciliation Act of 1993, which affected mostly top income earners; the Economic Growth

[49] Same as RAND: married and married with "spouse absent" as defined in the HRS survey are both recorded as jointly married; observations are recorded as single otherwise.

[50] For 1992 and 1993, we record dependents as max(children ever, non-resident children). No information about dependents was found in the 1995 FAT File, so we extend values of the previous wave for all interviewees. For 1996, we use the resident children variable instead as the dependents variable is not available. For all other waves, we use the number of dependents variable.

[51] Only available starting at wave 3.

[52] Same as RAND. H1CHKIN and H2CHKIN are not available, so we use H1ISAV1 and H2ISAV2 instead. Unlike HwCHKIN variables, which document both savings and checking interest income, H1ISAV1 and H2ISAV2 only record savings interest.

[53] Same as RAND except that Medicare part D) premiums are not added back (HRS only started recording it in 2006) and additional premiums deducted for higher income individuals, which started in 2007, are not added back. Medicare part B) coverage variable is not available for 1994.

[54] Values exceeding 999996 are reset to zero. These represent "don't know"/"refused to answer" responses to the survey question instead of an actual dollar amount.

[55] Values exceeding 999994 are reset to zero. These represent "don't know"/"refused to answer" responses to the survey question instead of an actual dollar amount.

[56] The medical expenditure variable is not available for wave 1. Except for wave 2 variable of medical expenditure, which records only one year of medical expense, we divide the observation by 2 to reflect that the medical expenditure is over a two-year period. We follow all of RAND's procedure; we interpret worker AGI as the sum of the AGIs of the primary respondent and spouse.

[57] Like RAND, we use the 30-year fixed rate mortgages (FRMs) multiplied by HwAMORT to calculate mortgage interest. However, we use FRMs published by the Federal Reserve Bank of St. Louis instead and average weekly values for the whole year to find the interest rate for the corresponding year. For the dollar amount of donations variable, values exceeding 999995 are reset to zero. These represent "don't know"/"refused to answer" responses to the survey question instead of an actual dollar amount. Again, in calculating medical expenditure that is not a preference for AMT, the medical expenditure variable is not available for wave 1. Except for wave 2 variable of medical expenditure, which records only one year of medical expense, we divide the observation by 2 to reflect that the medical expenditure is over a two-year period. We follow all of RAND's procedure, which now uses slightly different equations than the ones in computing for itemized deductions that are a preference for AMT; we interpret worker AGI as the sum of the AGIs of the primary respondent and spouse.

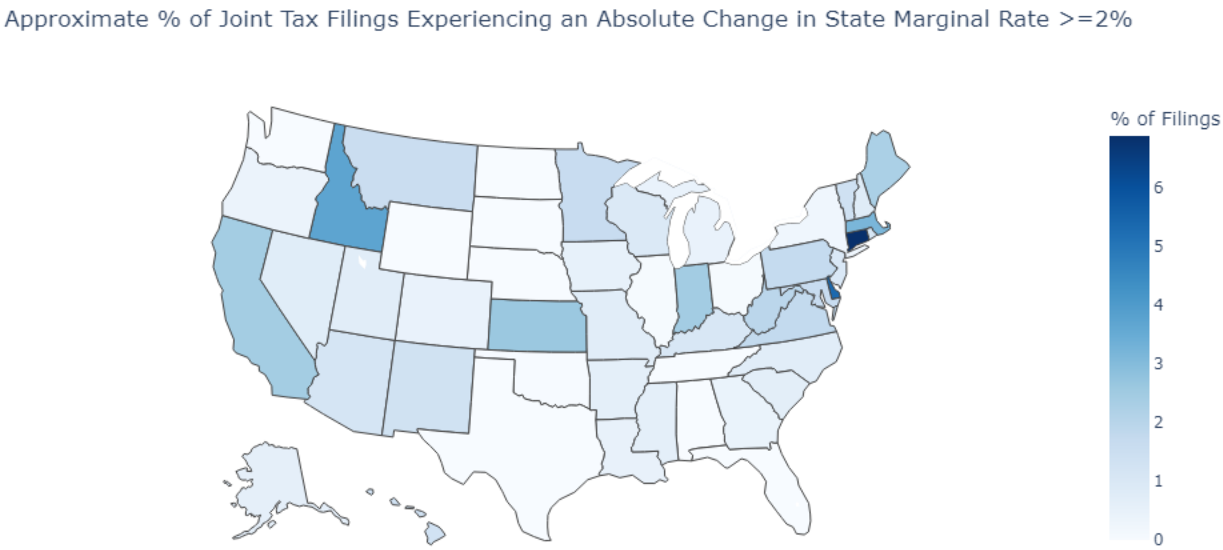
Table 3: TAXSIM and HRS data correspondence

	HRS Variable(s) Used	taxsim32 variable
1	N/A	taxsimid
2	RwIWENDY	year
3	RwSTATE	state
4	RwMSTAT[49]	mstat
5	RwAGEY_B	page
6	SwAGEY_B	sage
7	We use the dependents variable when available and impute values when needed.[50]	depx
8	N/A	dep13
9	N/A	dep17
10	N/A	dep18
11	RwIEARN	pwages
12	SwIEARN	swages
13	HwIDIVIN[51]	dividends
14	N/A (but this variable is not available in RAND's version of taxsim)	intrec
15	N/A	stcg
16	N/A	ltcg
17	HwRNTIN, HwIOTHI1, HwIOTHI2, HwITRSIN, HwIBNDIN, HwIBUSIN, HwICHKIN, HwIDIN, HwIALMNY, HwICDIN, HwICHKIN, H1ISAV1, H2ISAV2, HwIBUSIN, HwILUYR1-HwILUYR3[52]	otherprop
18	N/A	nonprop
19	RwIPENA, SwIPENA	pensions
20	RwISSDI, RwISSI, SwISSDI, SwISSI, Medicare part B) coverage variable from FAT Files[53]	gssi
21	RwIUNEM, SwIUNEM	ui
22	HwISSI, HwIFOOD, HwIWELF, RwIWCMP, RwIVET	transfers
23	Dollar amount of rent paid variable from FAT Files[54]	rent paid
24	Dollar amount of real estate tax paid variable from FAT Files[55]	proptax
25	RwOOPMD, SwOOPMD[56]	otheritem
26	N/A	childcare
27	HwAMORT, Dollar amount of donations variable from FAT Files, RwOOPMD, SwOOPMD[57]	mortgage
28	N/A	scorp
29	N/A	pbusinc
30	N/A	pprofinc
31	N/A	sbusinc
32	N/A	sprofinc

and Tax Relief Reconciliation Act of 2001, which affected those at the bottom and at top of the income distribution; the Jobs and Growth Tax Relief Reconciliation Act of 2003, which affected tax rates for middle and top income earners; the American Recovery and Reinvestment Act of 2009, with tax changes across different parts of the income distribution; and the American Taxpayer Relief Act of 2012, which changed tax rates at the top of the income distribution.

State tax reforms were spread across the US, as shown in Figure 10. They were more prevalent in some states such as California, Connecticut, Delaware, and Idaho, and were less frequent in Alaska, Florida, Nebraska, North Dakota, South Dakota, Tennessee, Texas, Washington, and Wyoming. The latter group includes states that have no state income taxes. Interestingly, both groups include states that index their tax brackets to inflation. To observe the changes in state income tax codes, we obtain income percentile cutoffs by converting all nominal incomes reported in the HRS RAND dataset to 2021 dollars, using the PCE deflator. Then, we construct a pseudo dataset with these cutoffs in all 50 states as well as Washington DC from 1992 to 2018. We then use the NBER tax simulator (taxsim32) to simulate marginal income tax rates for these constructed individuals. We then increment year by one, inflate accordingly using the PCE index, and obtain a new set of simualated marginal income tax rates. Finally, by taking the difference between the two rates, we find the policy changes at each income level.

Figure 10: State variation in marginal tax rates



Notes. Percentages were calculated using the NBER tax simulator and the publicly available HRS RAND longitudinal file, imputing each state distribution of income from the national distribution of income, and holding real incomes fixed across years. Shades are proportional to the percentage of jointly-filing taxpayers experiencing an absolute marginal tax change larger than 2%.

C.4 Mental Status Scores

The mental status summary sums the scores for serial 7's (RwSER7, 0-5), backwards counting from 20 (RwBWC2, 0-2), and object (RwCACT, RwSCIS; 0-2 total), date (RwDY, RwMO, RwYR, RwDW; 0-4 total), and President/Vice-President (RwPRES, RwVP; 0-2 total) naming tasks. The resulting range is 0-15. Since these items were not included in Waves 1 and 2H, there is no mental summary score for these waves, and the Wave 2A summary is called R2AMSTOT to indicate that it is limited to the AHEAD cohort in Wave 2.

Those questions are presented in the table below.

Table 5: Mental Status Scores - Questions

Variable	Content	Score
RwCACT	"What do you call the kind of prickly plant that grows in the desert?"	0-1
RwSCIS	"What do you usually use to cut paper?"	0-1
RwSCIS and RwVP	Whether the Respondent was able to correctly name the current president and vice-president of the United States, respectively.	0-2
RwSER7	Number of correct subtractions in the serial 7s test. This test asks the individual to subtract 7 from the prior number, beginning with 100 for five trials. Correct subtractions are based on the prior number given, so that even if one subtraction is incorrect subsequent trials are evaluated on the given (perhaps wrong) answer.	0-5
RwBWC20 and RwBWC86	Whether the Respondent was able to successfully count backwards for 10 subsequent numbers from 20 and 86, respectively. Two points are given if successful on the first try, one if successful on the second, and zero if not successful on either try.	0-2
RwDY, RwMO, RwYR, and RwDW	Whether the Respondent was able to report today's date correctly, including the day of month, month, year, and day of week, respectively.	0-4